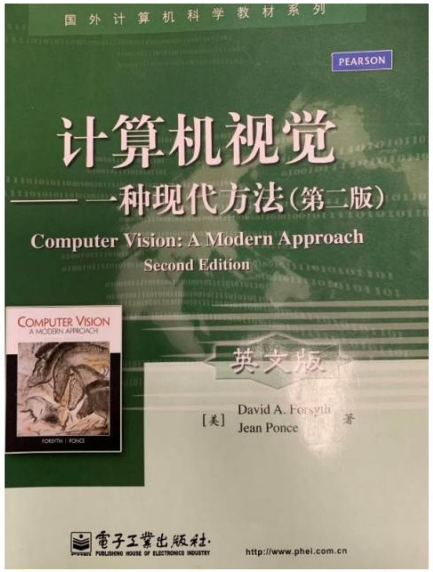
Computer Vision 计算机视觉

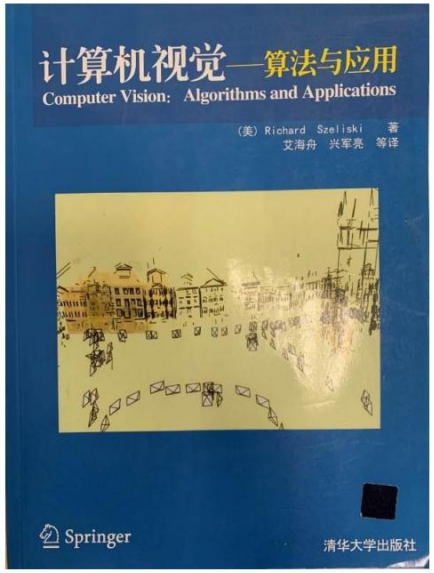
Lecture 1 第 1 讲

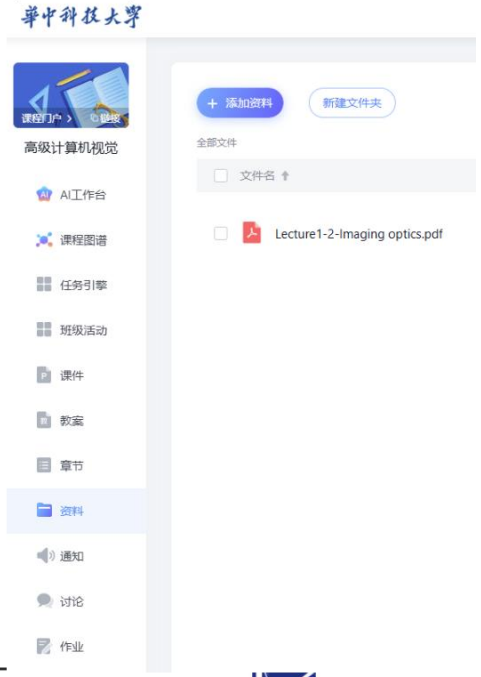
Light and Color 光与颜色

About the course 课程概述

- 教师：桑农，王岳环，左峥嵘
- 教材及参考资料 Computer Vision《计算机视觉》 selected papers 选读论文
- 成绩构成: 闭卷考试（课程结束后2周）70%+平时成绩30%（按照讲授内容写一份心得，包括对课程内容的理解，存在的问题和难点，对讲授内容的需求等。）
- 日常通知及课件: 课程平台
- 课程选定后不得无故修改，必须参加课程学习考核，如因特殊原因需要退课，必须在课程进度的前50%内申请，逾期不予处理；







About the course 课程概述

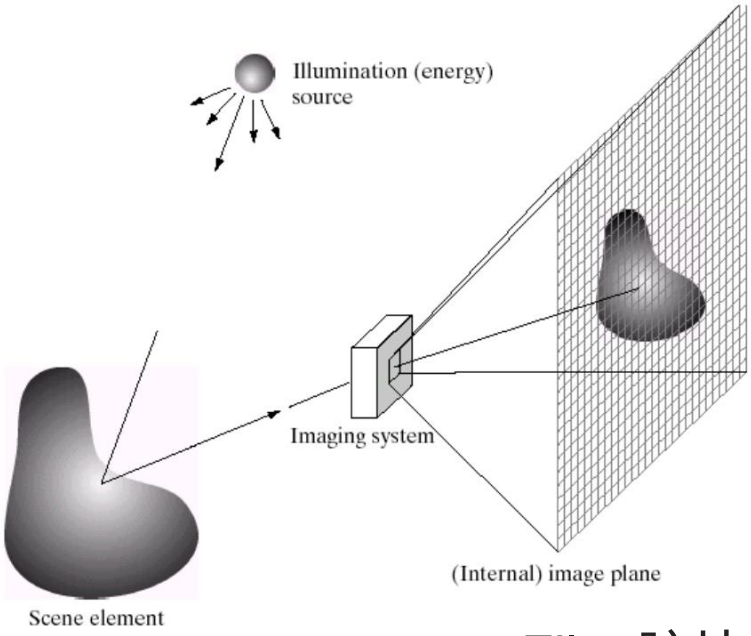
What is computer vision?

计算机视觉的研究内容

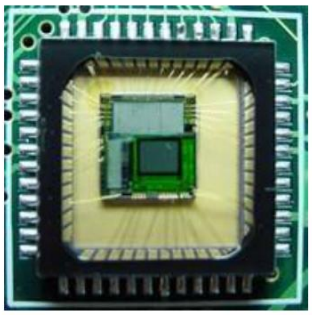
Where the ideas come from?

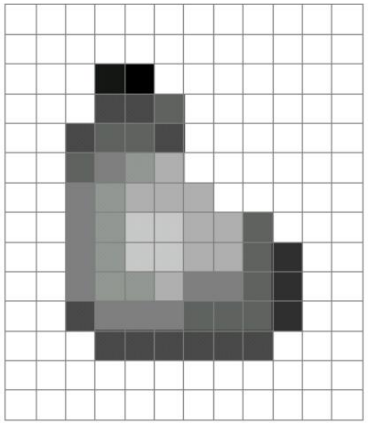
计算机视觉的思想来源

Image Formation 图像形成

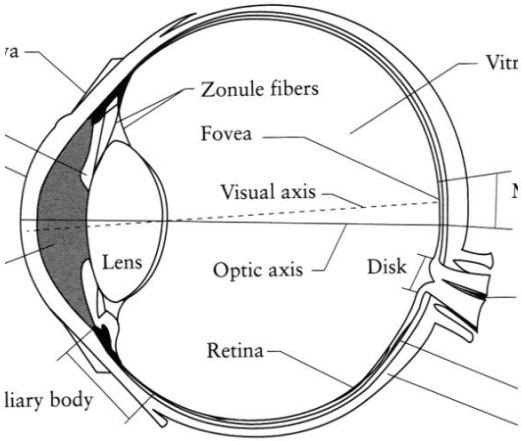


Film 胶片





Digital Camera 数码相机

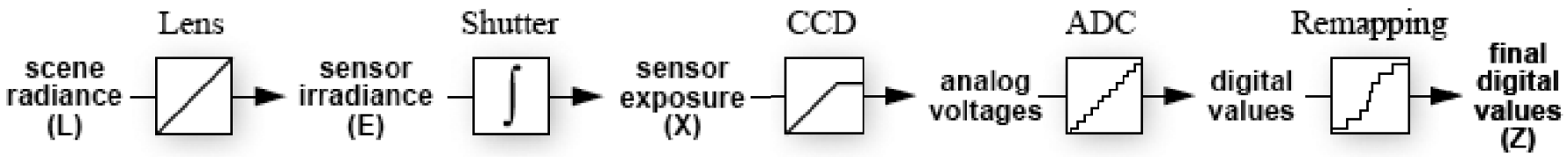


The Eye 人眼

Major factors 主要影响因素

Illumination strength and direction	光照强度与方向
Surface geometry	表面几何形状
Surface material	表面材质
Nearby surfaces	周围表面
Camera gain/exposure	相机增益与曝光

From light rays to pixel values 光线到像素值的转换



Lens 镜头　　Shutter 快门　　CCD 电荷耦合器件　　ADC 模数转换　　Remapping 重映射

L 场景辐亮度　　E 传感器辐照度　　X 传感器曝光量　　Z 最终数字值

analog voltages 模拟电压　　digital values 数字值

For more information 延伸阅读

M. Brown, Understanding the In-Camera Image Processing Pipeline for Computer Vision, CVPR 2016 Tutorial

计算机视觉中的相机内部图像处理流程

A photon's life choices 光与物质的相互作用

Absorption 吸收

Diffusion 漫射

Reflection 反射

Transparency 透射

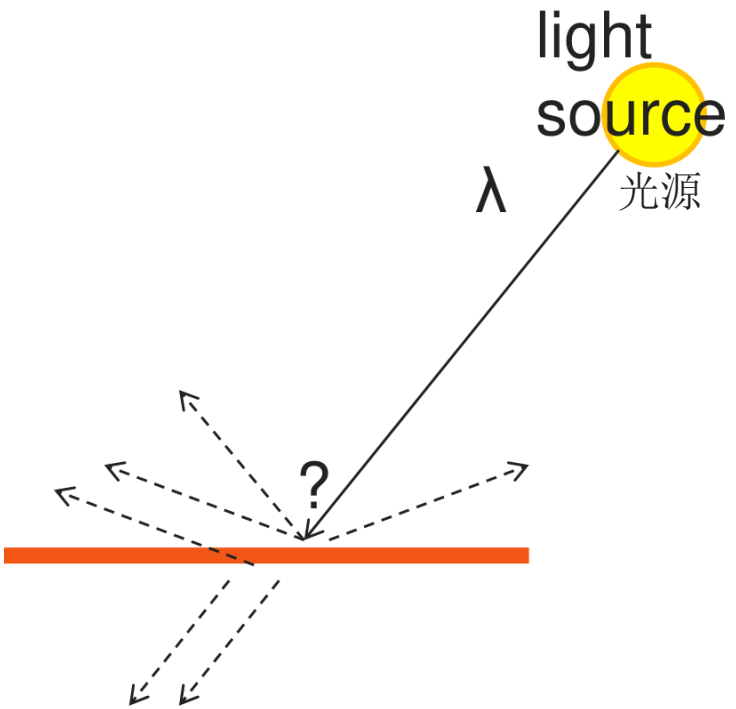
Refraction 折射

Fluorescence 荧光

Subsurface scattering 次表面散射

Phosphorescence 磷光

Interreflection 表面间反射



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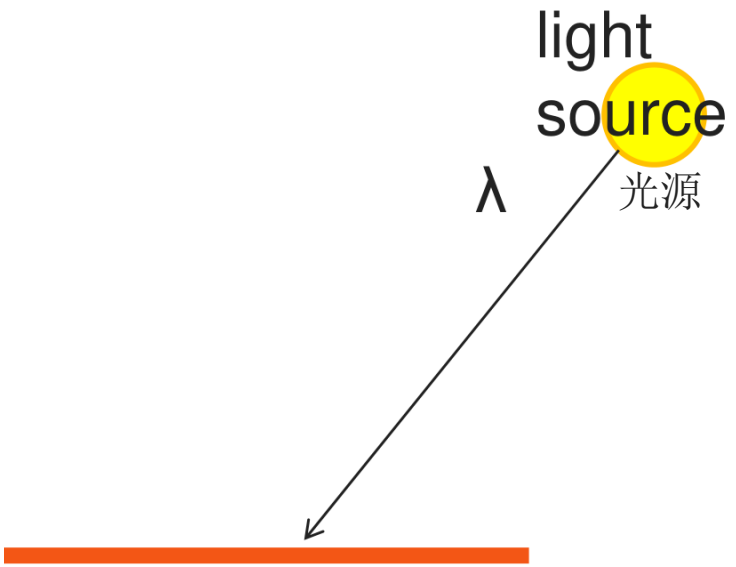

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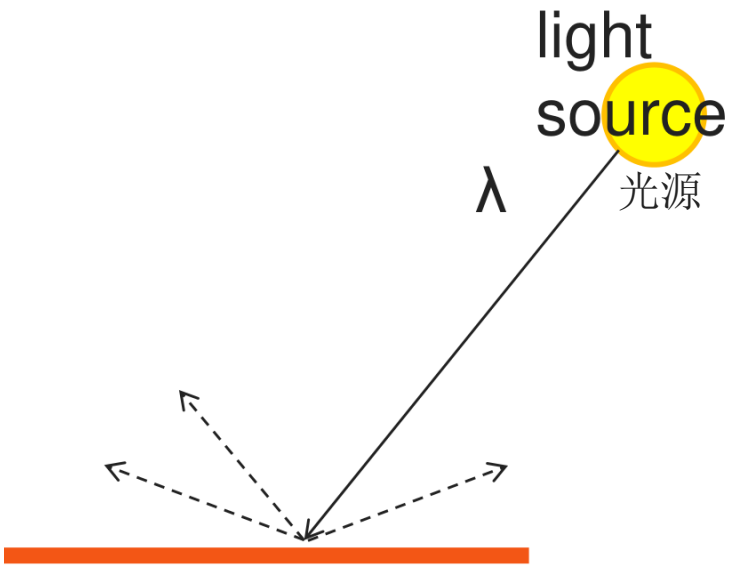

A photon's life choices 光与物质的相互作用

Absorption 吸收

Diffusion 漫射

Specular Reflection 镜面反射

Transparency 透射

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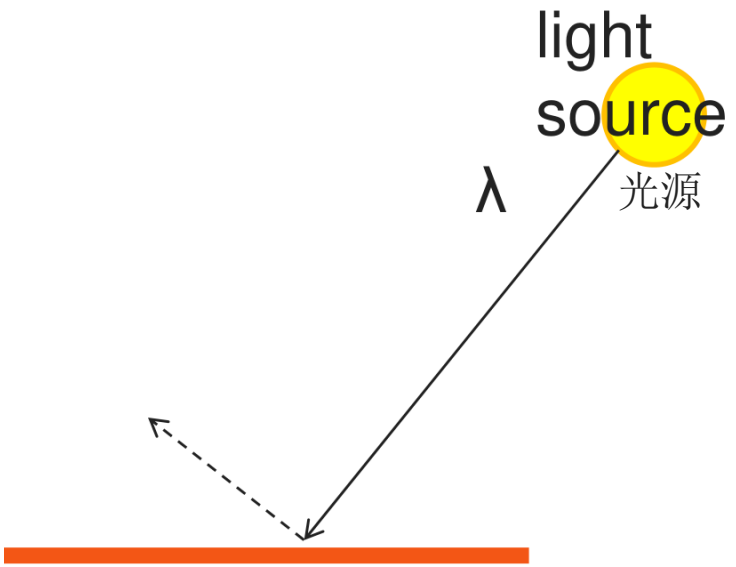

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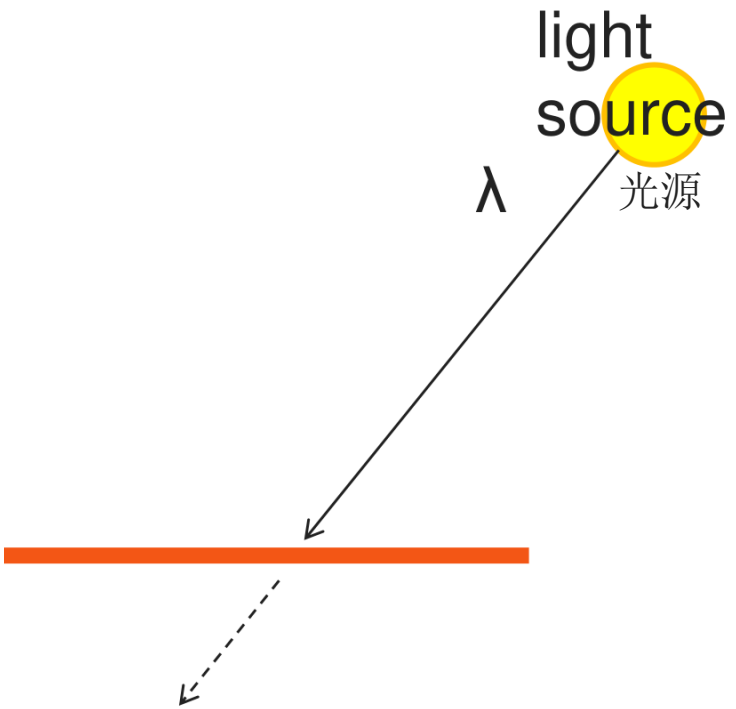

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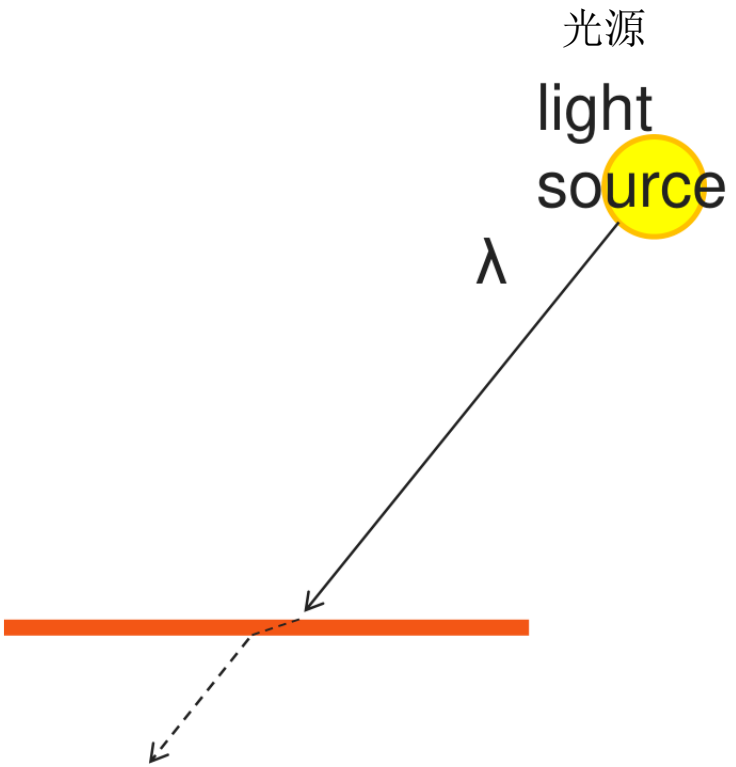

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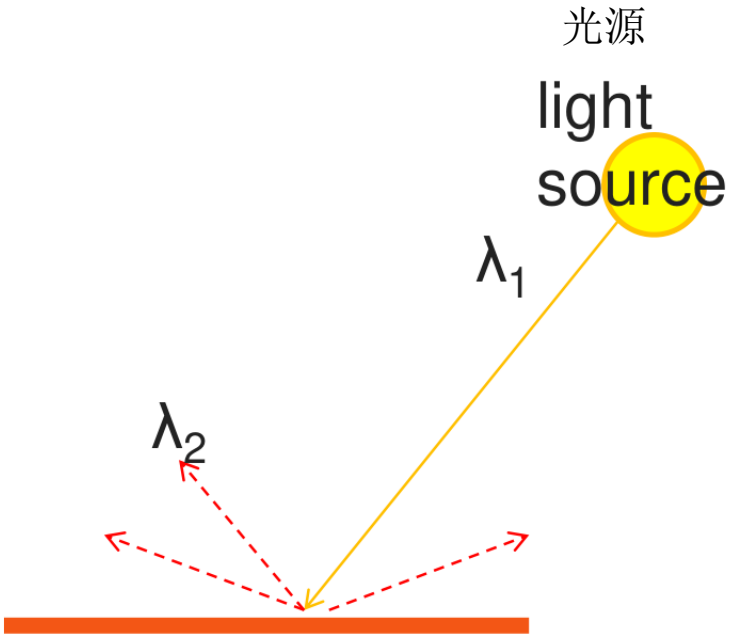

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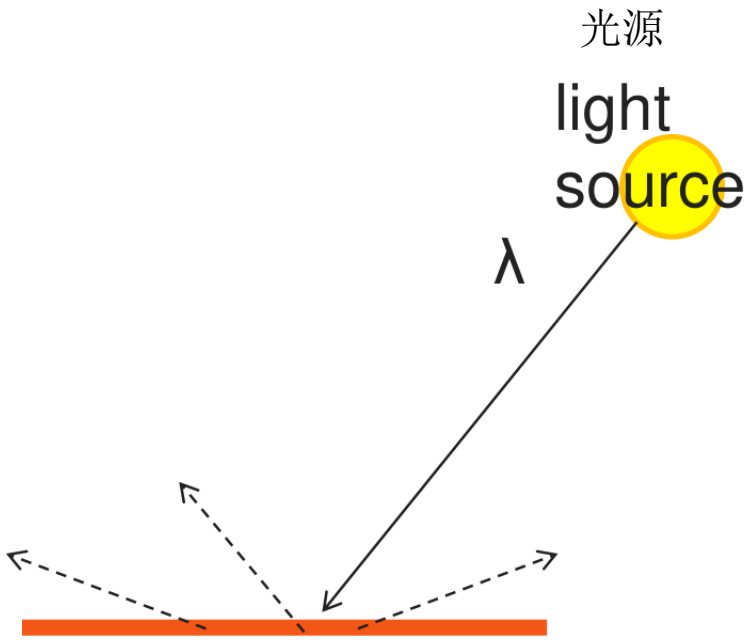

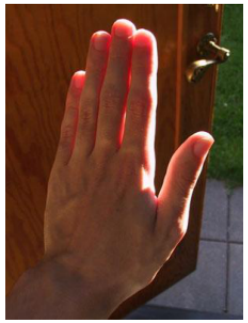

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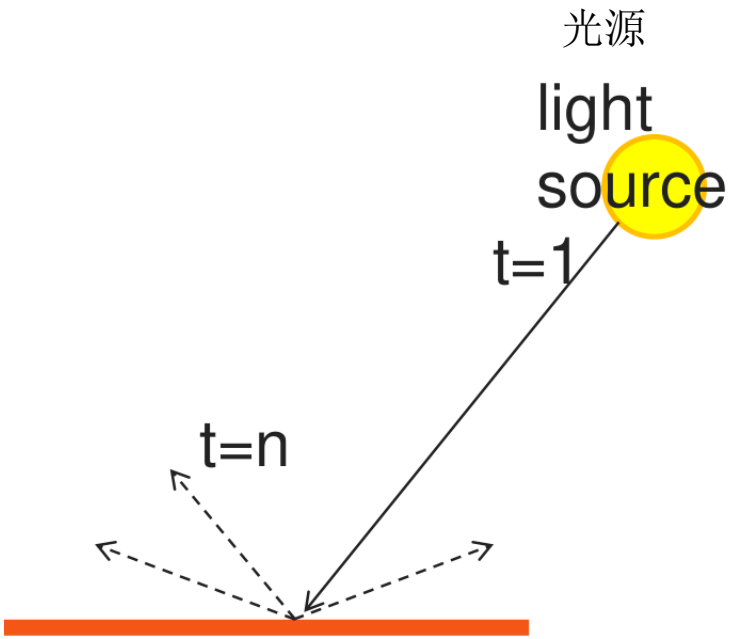

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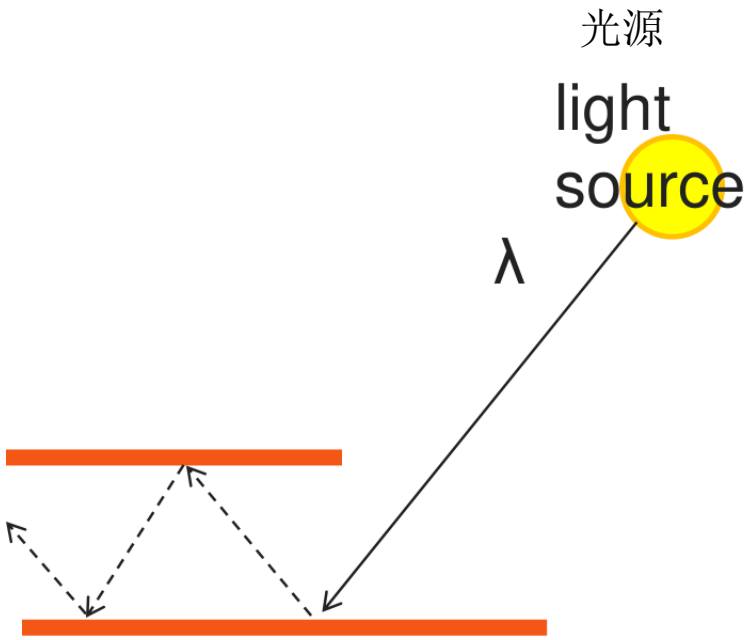


(Specular Interreflection)

镜面表面间反射

Radiometry 辐射度学

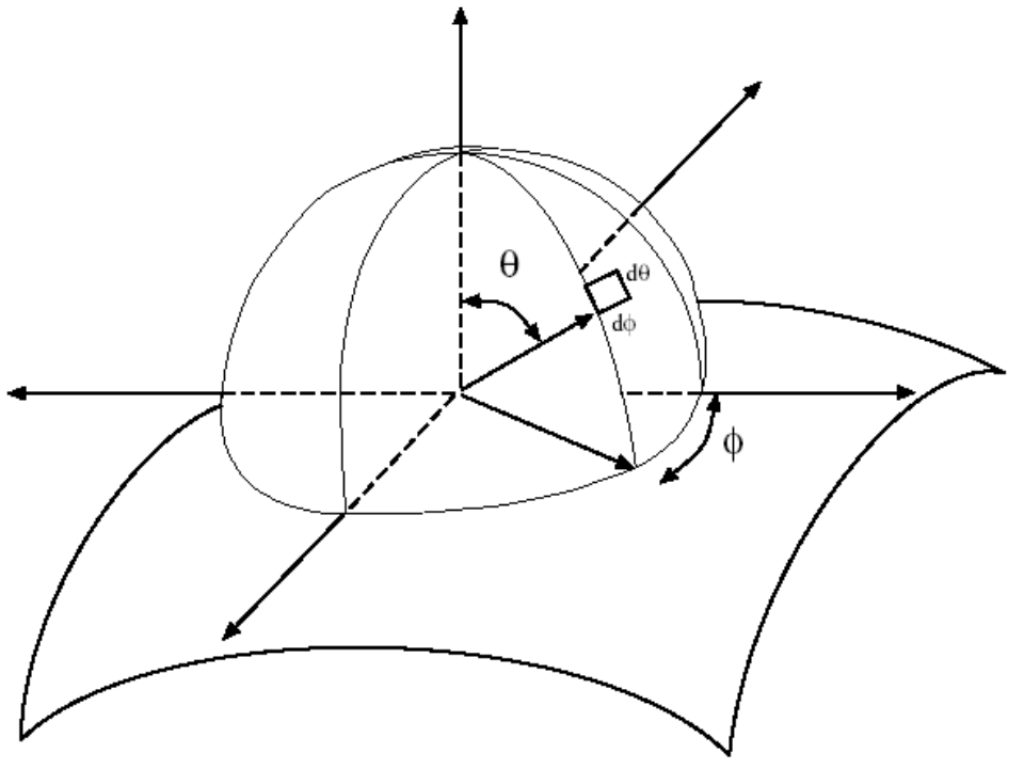
- *Radiometry* is the part of image formation concerned with the relation among the amounts of
  - light energy emitted from light sources,
  - reflected from surfaces,
  - and registered by sensors.

辐射度学属于图像形成的研究内容，研究以下各类光能量之间的数量关系。
光源发出的光能量
表面反射的光能量
传感器记录的光能量

Foreshortening 投影缩短

- A point on a surface sees the world along a hemisphere of directions centered at the point.

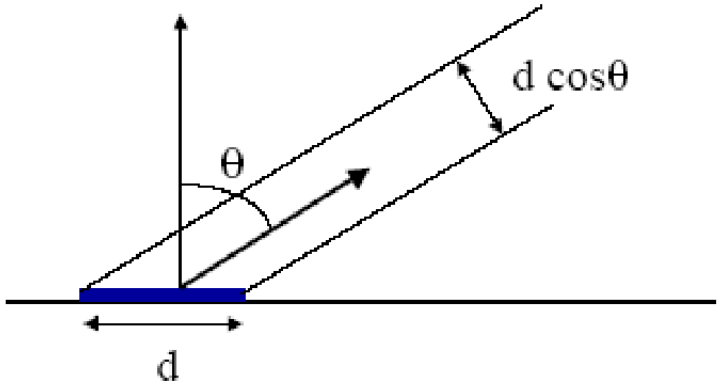
  表面上的一个点，可沿以该点为中心的半球内各个方向观察周围世界。



Foreshortening 投影缩短

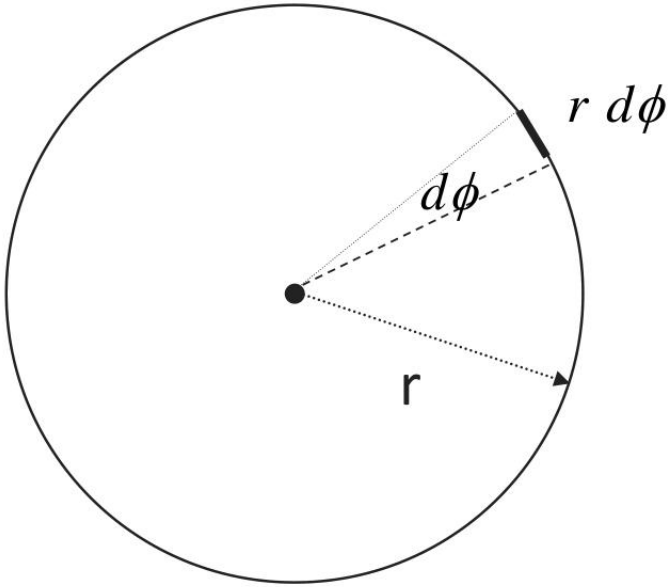
- A big source, viewed at a glancing angle, must produce the same effect as a small source viewed frontally.
- This phenomenon is known as *foreshortening*.

以掠射角观察较大光源时，其效果应与正面观察较小光源时相同。
这种现象称为投影缩短。



Solid Angle 立体角

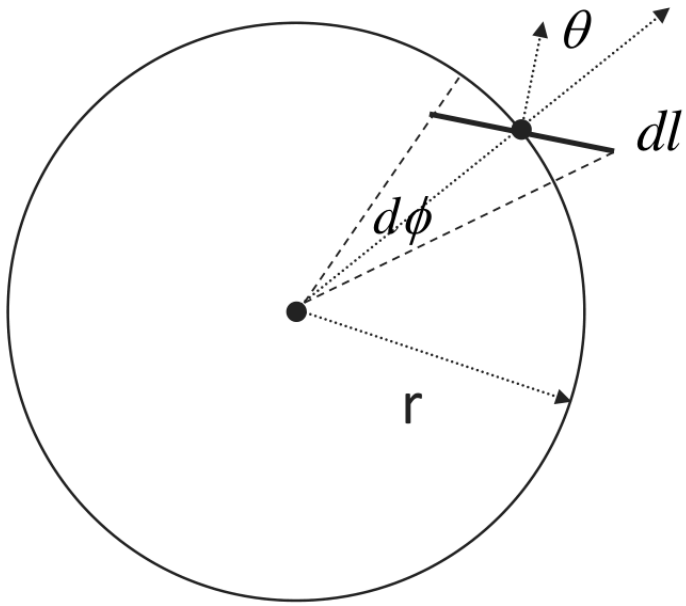
- Arc length 弧长



Solid Angle 立体角

- Similarly, solid angle due to a line segment is

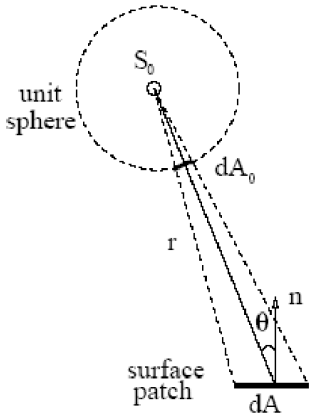
  类似地，线段张成的角度可由下式表示。



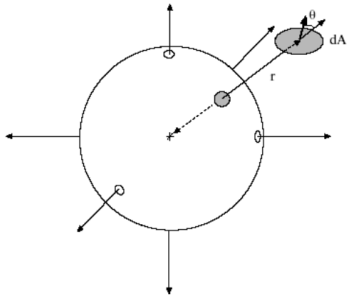
$$d\phi = \frac{dl\cos\theta}{r}$$

Solid Angle 立体角

- *Solid angle* is defined by the projected area of a surface patch onto a unit sphere of a point.



$$d\omega = dA_0 = \frac{dA\cos\theta}{r^2}$$



(Solid angle is subtended by a point and a surface patch.)

立体角由表面微元投影到以观察点为球心的单位球面上的面积定义。

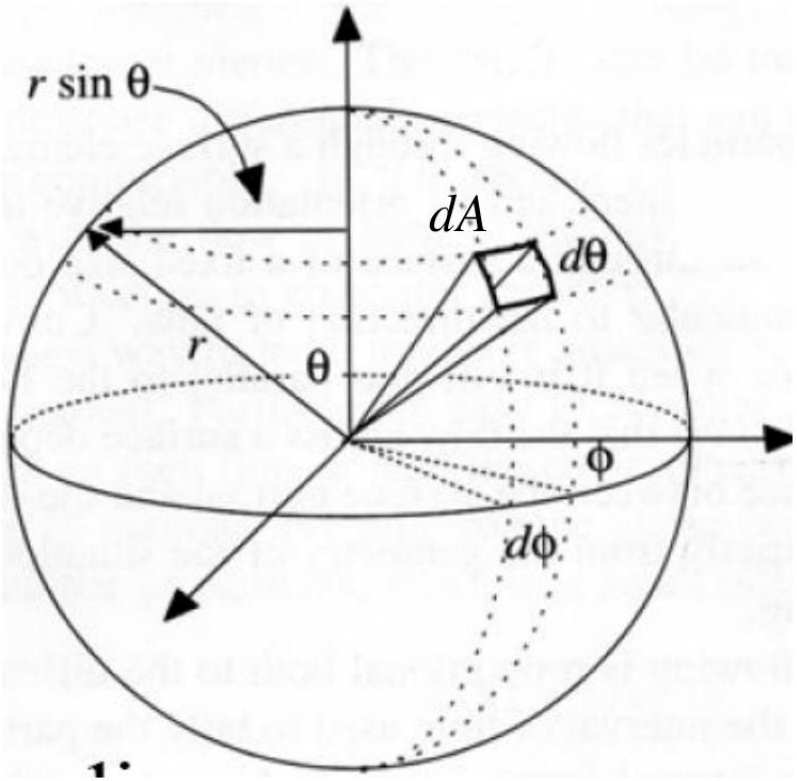
一个点与一个表面微元共同确定所张的立体角。

unit sphere 为单位球面，surface patch 为表面微元。

Solid Angle 立体角

- *Solid angle* is defined by the projected area of a surface patch onto a unit sphere of a point.

  立体角由表面微元投影到以观察点为球心的单位球面上的面积定义。



$$dA = rd\theta r\sin\theta d\phi = r^2\sin\theta d\theta d\phi$$

$$TotalArea = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi} r^2\sin\theta d\theta d\phi = 4\pi r^2$$

TotalArea 为球面总面积

$$dw = \frac{dA}{r^2} = \sin\theta d\theta d\phi$$

Radiance 辐亮度

- The distribution of light in space is a function of position and direction.
- The appropriate unit for measuring the distribution of light in space is *radiance*, which is defined as the *power* (the amount of energy per unit time) traveling at some point in a specified direction, per unit area *perpendicular to the direction of travel*, per unit solid angle.
- In short, radiance is the amount of light radiated from a point… (into a unit solid angle, from a unit area).

Radiance = Power / (solid angle x foreshortened area)

W/sr/m2
W is Watt, sr is steradian, m2 is meter-squared

光在空间中的分布是位置与方向的函数。

辐亮度适于度量光在空间中的分布，其定义为光在某点沿指定方向传播时，单位垂直投影面积、单位立体角内的功率。功率是单位时间内的能量。

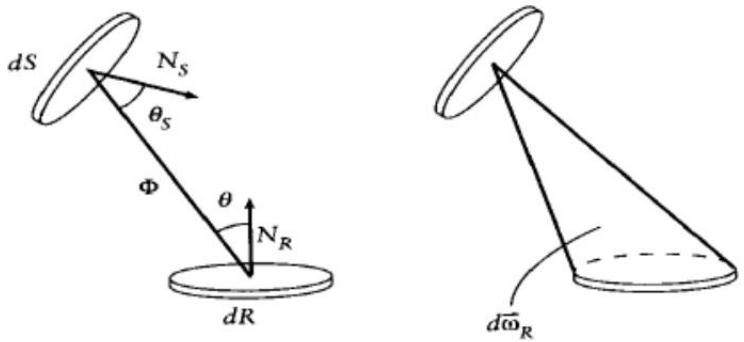
辐亮度描述从单位面积向单位立体角辐射的光量。

辐亮度 = 功率 /（立体角 × 投影面积）

单位为 W/sr/m²。W 表示瓦特，sr 表示球面度，m² 表示平方米。

Radiance 辐亮度

- Radiance from dS to dR 从 dS 到 dR 的辐亮度



*Radiance = Power / (solid angle * foreshortened area)*

辐亮度 = 功率 /（立体角 × 投影面积）

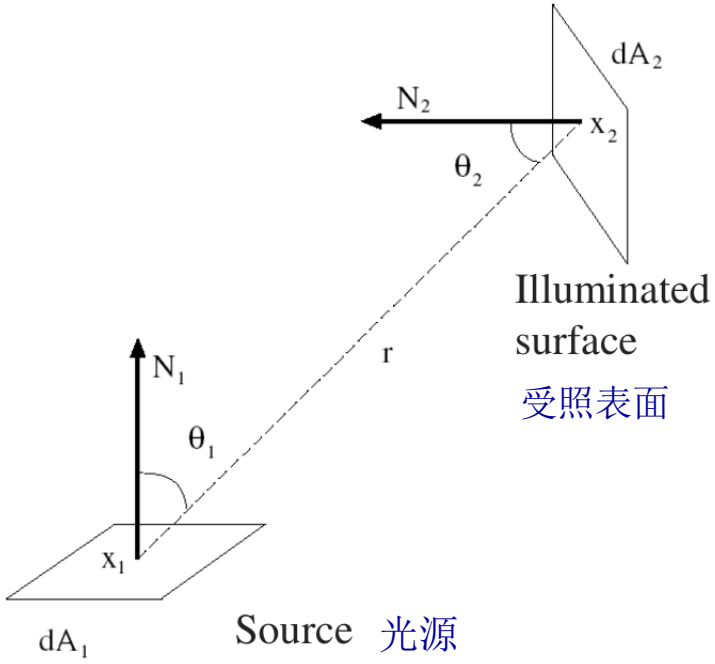
Radiance 辐亮度

- Example: Infinitesimal source and surface patches

  示例 无穷小光源与表面微元

  Radiance = Power / (solid angle * foreshortened area)

  辐亮度 = 功率 /（立体角 × 投影面积）



从 x_1 沿 x_2 方向出射的辐亮度

Radiance at x1 leaving to x2

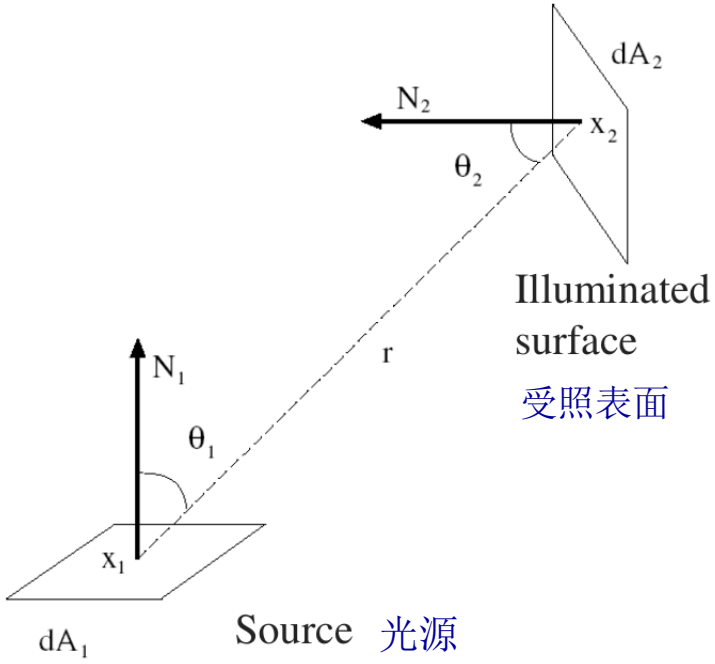
$$L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_1 dA_1} = \frac{r^2 d\psi}{dA_2 \cos\theta_2 \cos\theta_1 dA_1}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance 辐亮度

Radiance = Power / (solid angle * foreshortened area)

辐亮度 = 功率 /（立体角 × 投影面积）



从 x_1 沿 x_2 方向出射的功率

Power at x1 leaving to x2

$$d\psi = L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2)\, dw \cos\theta_1 dA_1$$

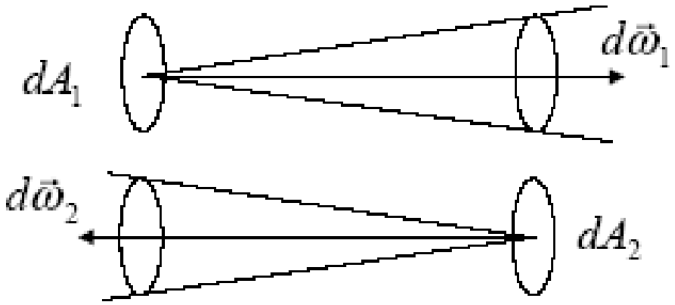
$$= \frac{L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2)\, dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Radiance 辐亮度

- If the medium is vacuum, power is preserved.

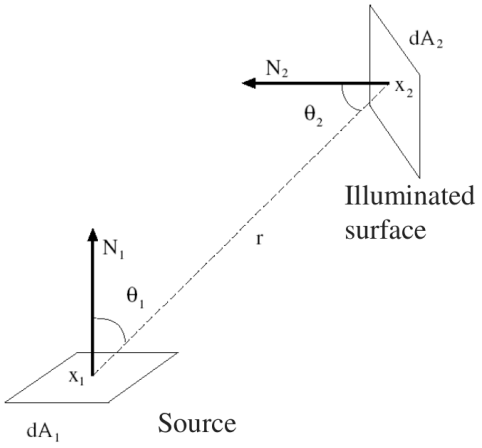
  当传播介质为真空时，光功率守恒。



$$L_1 d\vec{\omega}_1 dA_1 = L_2 d\vec{\omega}_2 dA_2$$

Radiance 辐亮度

- The medium is vacuum, that is, it does not absorb energy. Therefore, the power reaching point x2 is equal to the power leaving for x2 from x1.



Power at x2 from direction x1 is

$$d\psi = \frac{L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) dA_2 \cos\theta_2 \cos\theta_1 dA_1}{r^2}$$

Let the radiance arriving at x2 from the direction of x1 is

$$L(\mathbf{x}_2, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = \frac{d\psi}{dw \cos\theta_2 dA_2} = \frac{r^2 d\psi}{dA_1 \cos\theta_1 \cos\theta_2 dA_2}$$

$$dw = \frac{dA_1 \cos\theta_1}{r^2}$$

传播介质为真空，能量吸收为零。因此，到达点 x_2 的光功率，等于从点 x_1 向点 x_2 发出的光功率。

从点 x_1 方向到达点 x_2 的功率由上式给出。

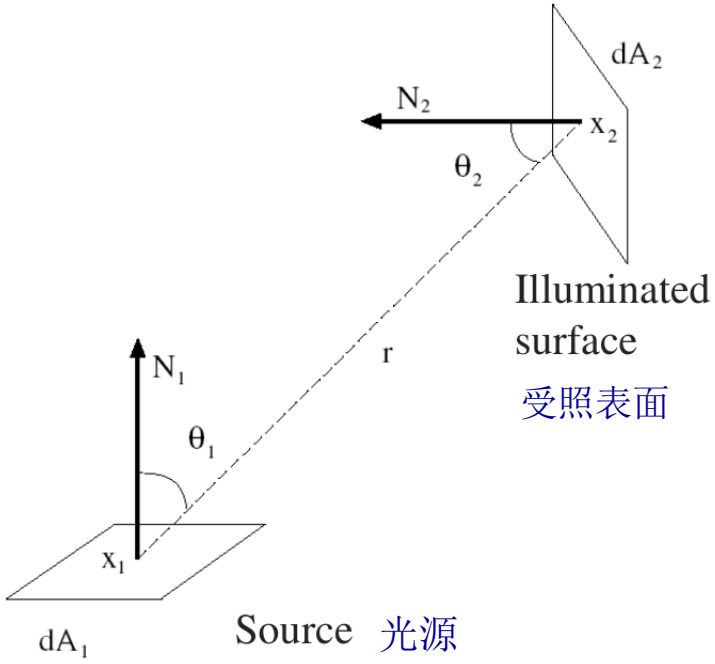
设从点 x_1 方向到达点 x_2 的辐亮度为下式。

Source 为光源。
Illuminated surface 为受照表面。

Radiance 辐亮度

- **Radiance is constant along a straight line.**

  辐亮度沿直线传播时保持恒定。



$$L(\mathbf{x}_1, \mathbf{x}_1 \rightarrow \mathbf{x}_2) = L(\mathbf{x}_2, \mathbf{x}_1 \rightarrow \mathbf{x}_2)$$

Point Source 点光源

- Many light sources are physically small compared with the environment in which they stand.
- Such a light source is approximated as an extremely small sphere, in fact, a point.
- Such a light source is known as a *point source*.

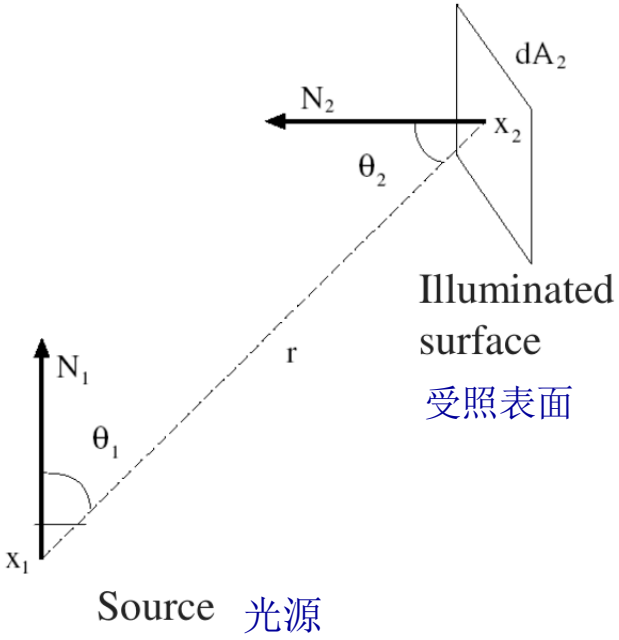
许多光源的实际尺寸相对于其所在环境很小。
这类光源可近似为尺寸极小的球体，即一个点。
这样的光源称为点光源。

Radiance Intensity 辐射强度

- If the source is a *point source*, we use radiance intensity.

  对于点光源，采用辐射强度描述其发光特性。

Radiance intensity = Power / (solid angle)



辐射强度 = 功率 / 立体角

$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_2 \cos\theta_2}$$

$$dw = \frac{dA_2 \cos\theta_2}{r^2}$$

Light at Surfaces 表面处的光

- When light strikes a surface, it may be absorbed, transmitted, or scattered; usually, combination of these effects occur.
- It is common to assume that all effects are local and can be explained with a **local interaction model**. In this model:
  - The radiance leaving a point on a surface is due only to radiance arriving at this point.
  - Surfaces do not generate light internally and treat sources separately.
  - Light leaving a surface at a given wavelength is due to light arriving at that wavelength.

光照射到表面时，可能发生吸收、透射或散射，通常会同时出现这些效应。

通常假设所有效应均发生于局部，并用局部相互作用模型加以解释。该模型具有以下假设。

表面某点的出射辐亮度完全由到达该点的入射辐亮度决定。

表面内部的发光量视为零，光源另行处理。

表面在某一波长的出射光，源于同一波长的入射光。

Light at Surfaces 表面处的光

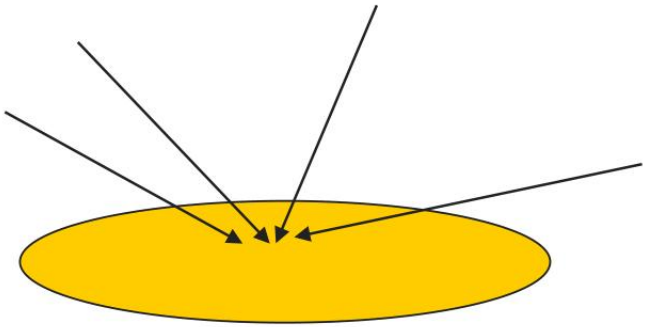
- In the local interaction model, *fluorescence*, [absorb light at one wavelength and then radiate light at a different wavelength], and *emission* [e.g., warm surfaces emits light in the visible range] are neglected.

局部相互作用模型忽略荧光和表面自身发光。
荧光指吸收某一波长的光后，发出另一波长的光。
表面自身发光的例子是受热表面发出可见光。

Irradiance 辐照度

- *Irradiance* is the total incident power per unit area.

  辐照度是单位面积上的总入射功率。



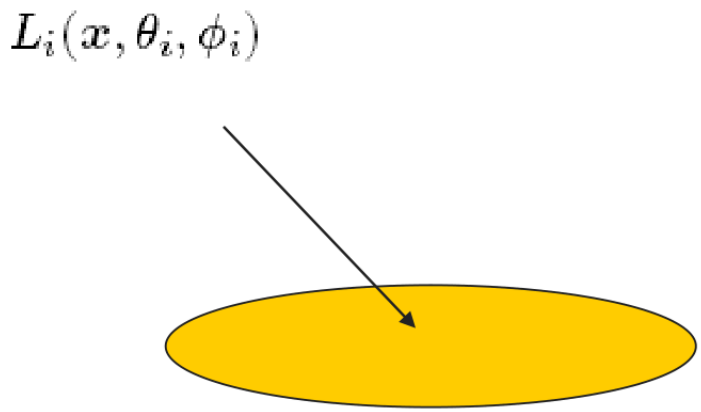
Irradiance = Power / Area

辐照度 = 功率 / 面积

Irradiance 辐照度

- What is the irradiance due to source from angle (θ_i, ϕ_i)?

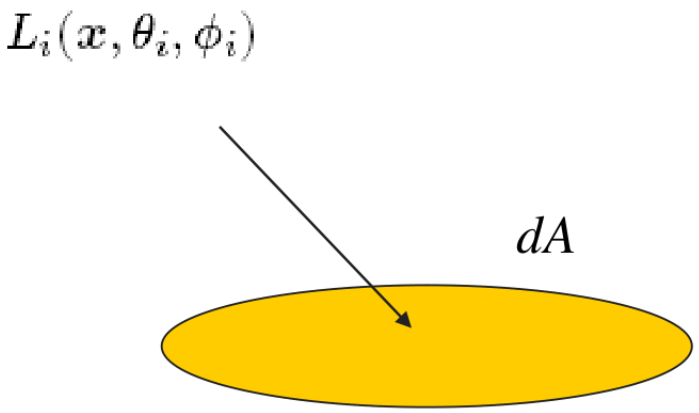
  来自方向（θ_i，φ_i）的光源所产生的辐照度



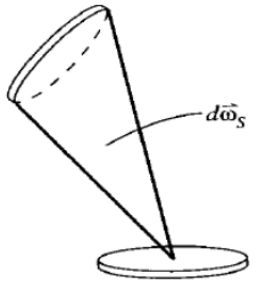
Irradiance 辐照度

- What is the irradiance due to source from angle (θ_i, ϕ_i)?

  来自方向（θ_i，φ_i）的光源所产生的辐照度



$$Irradiance = \frac{d\psi}{dA} = \frac{L_i(x,\theta_i,\phi_i)dw\cos\theta_i dA}{dA} = L_i(x,\theta_i,\phi_i)dw\cos\theta_i$$



辐照度 = 单位面积上的入射功率

Irradiance due to a Point Source　点光源产生的辐照度

- For a point source,　点光源情形

$$\text{Radiance_Intensity} = \frac{\text{power}}{\text{solid_angle_source}}$$

$$I = \frac{d\psi}{dw} = \frac{r^2 d\psi}{dA_i \cos\theta_i}$$

$$dw = \frac{dA_i \cos\theta_i}{r^2}$$

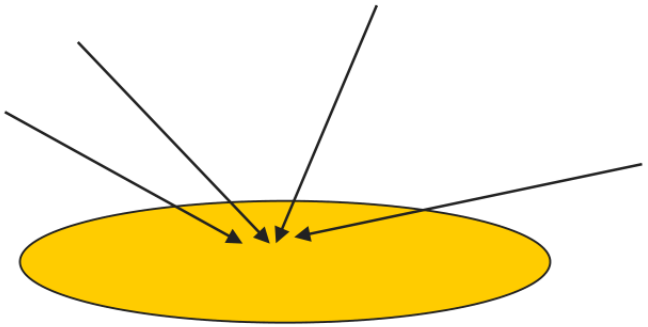
$$\Rightarrow \quad d\psi = I \frac{dA_i \cos\theta_i}{r^2} \quad \Rightarrow \quad Irradiance = \frac{d\psi}{dA_i} = I \frac{\cos\theta_i}{r^2}$$

辐射强度 = 功率 / 光源处的出射立体角

辐照度 = I $\cos\theta_i$ / r^2

Irradiance 辐照度

- What is the total irradiance? 总辐照度的计算



练习
假设各个方向的辐亮度均相同，计算辐照度。

Integrate over the whole hemisphere.

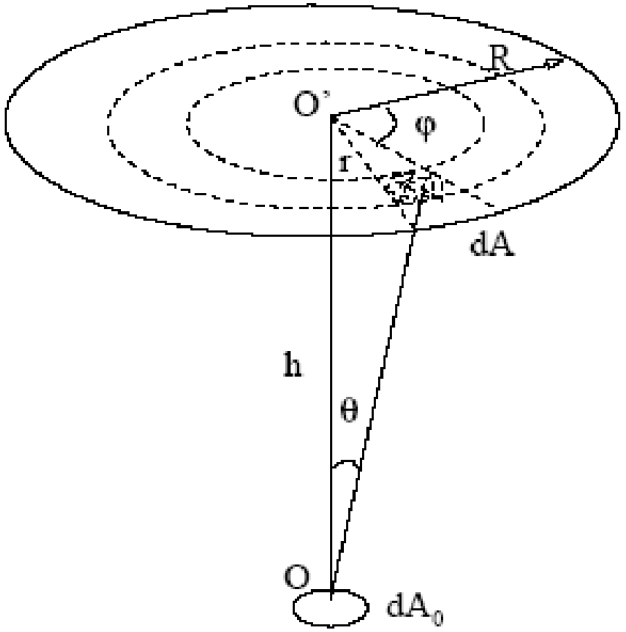
对整个半球积分。

Exercise: Suppose the radiance is constant from all directions. Calculate the irradiance.

Irradiance 辐照度

- Exercise: Calculate the irradiance at O due to a plate source at O'.

  练习　计算位于 O′ 的平面光源在 O 点产生的辐照度。



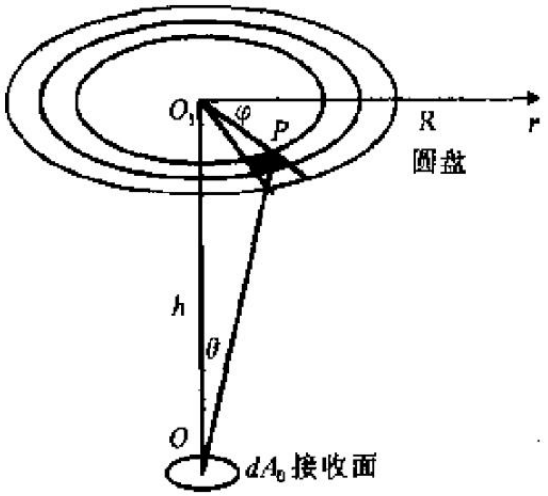
Irradiance due to a Surface Source　面光源产生的辐照度

$$d\omega = \frac{dA_0 \cos\theta}{(PO)^2}$$

$$d^2P = LdA.d\omega = L.r\cos\theta dr d\varphi \bullet \frac{dA_0 \cos\theta}{(PO)^2}$$

$$dP = \int_0^R \int_0^{2\pi} d^2P = L\frac{\pi R^2}{h^2 + R^2} dA_0$$

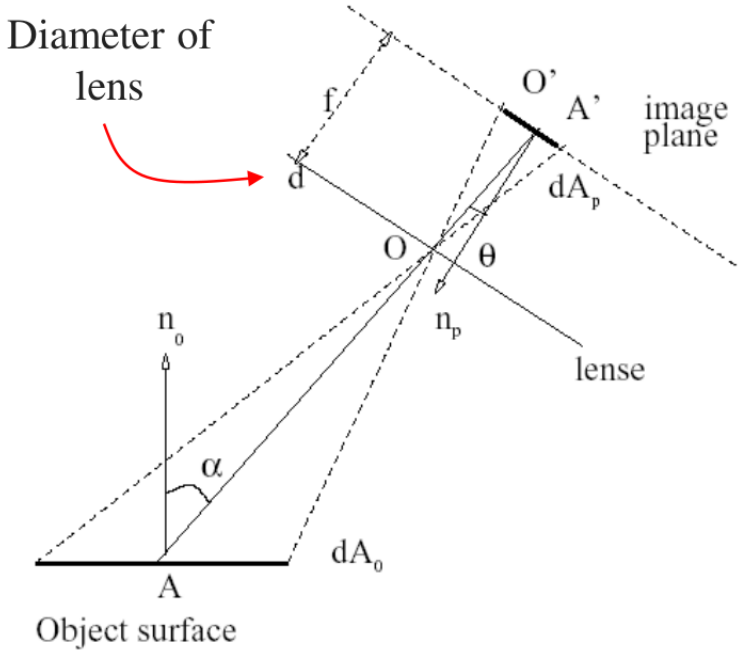
$$E = \frac{dP}{dA_0} = L\frac{\pi R^2}{h^2 + R^2}$$



$$\cos\theta = \frac{h}{PO}$$

The Relationship Between Image Intensity and Object Radiance
图像强度与物体辐亮度的关系

d 镜头直径　lense 镜头
image plane 像平面
object surface 物体表面



假设镜头中的光功率损耗为零。

We assume that there is no power loss in the lens.

发射到镜头的功率由下式给出。

The power emitted to the lens is

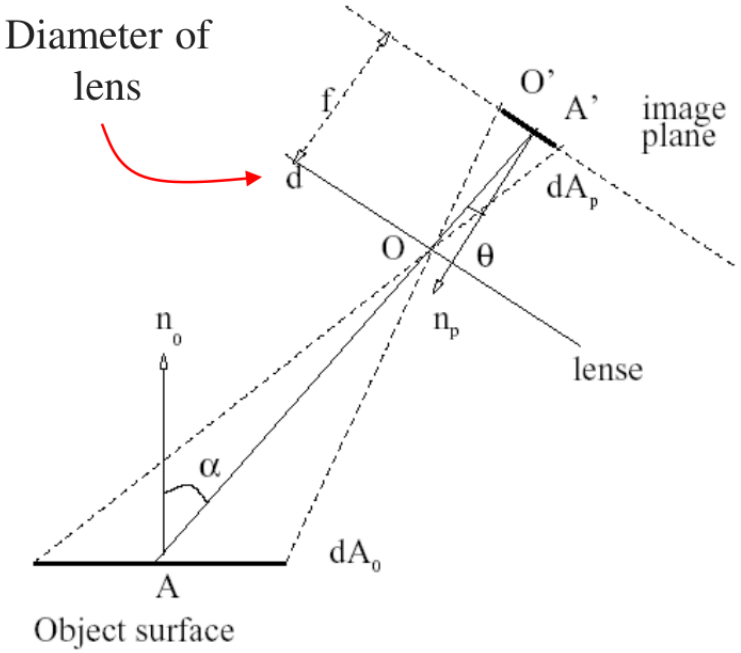
$$d\psi = L_{object} dA_0 \cos\alpha dw_0$$

Radiance of object

L_object 为物体辐亮度

The Relationship Between Image Intensity and Object Radiance
图像强度与物体辐亮度的关系

d 镜头直径　lense 镜头
image plane 像平面
object surface 物体表面



整个镜头所张的立体角由下式给出。

The solid angle for the entire lens is

$$dw_0 = \frac{\left(\pi d^2/4\right)\cos\theta}{r^2}$$

The power emitted to the lens is

$$d\psi = L_{object} dA_0 \cos\alpha dw_0$$

$$= L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

发射到镜头的功率由上式给出。

The Relationship Between Image Intensity and Object Radiance
图像强度与物体辐亮度的关系

d 镜头直径　lense 镜头
image plane 像平面
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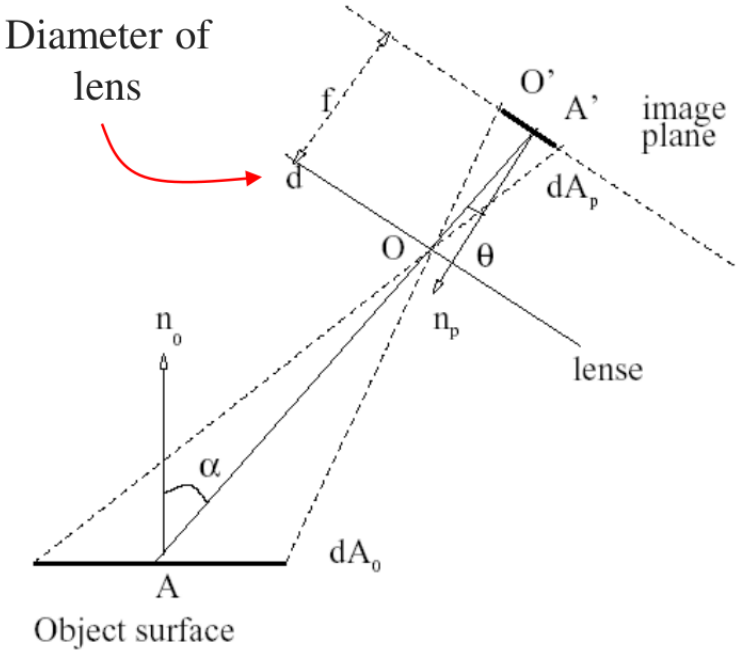


O 点处的立体角可写成两种形式。

The solid angle at O can be written in two ways.

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos\theta}{|OA'|^2}$$

Note that　利用下式

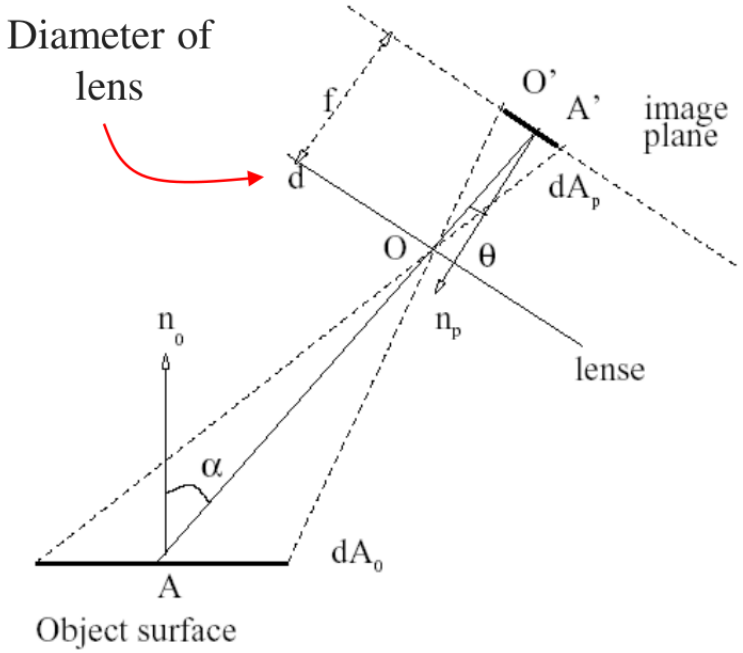
$$|OA'| = f / \cos\theta$$

Therefore　因此可得

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

The Relationship Between Image Intensity and Object Radiance
图像强度与物体辐亮度的关系

d 镜头直径　lense 镜头
image plane 像平面
object surface 物体表面



Combine　联立两式

$$\frac{dA_0 \cos\alpha}{r^2} = \frac{dA_p \cos^3\theta}{f^2}$$

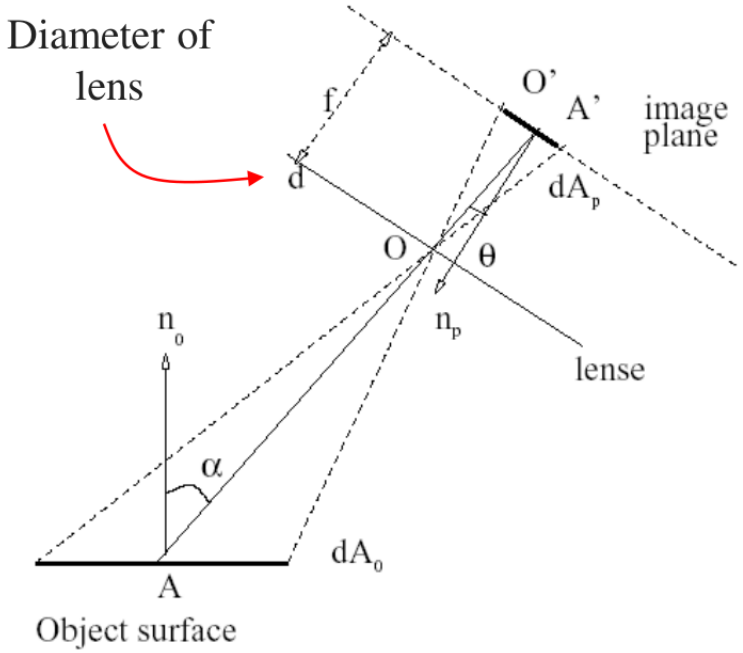
$$d\psi = L_{object} dA_0 \cos\alpha \frac{\pi d^2 \cos\theta}{4r^2}$$

to get　可得下式

$$d\psi = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta dA_p$$

The Relationship Between Image Intensity and Object Radiance
图像强度与物体辐亮度的关系

d 镜头直径　lense 镜头
image plane 像平面
object surface 物体表面



辐照度经转换得到像素强度，像素强度与物体辐亮度成正比。

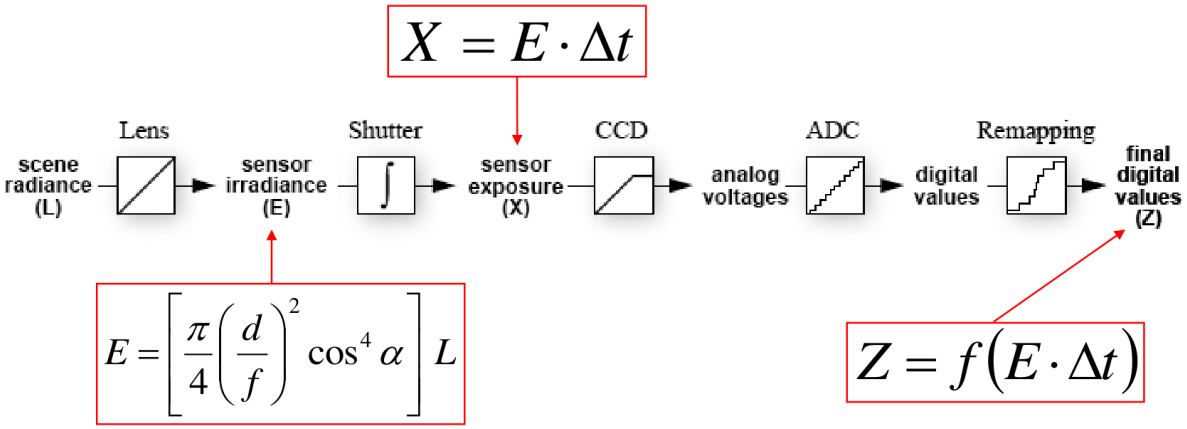
因此，像平面上的辐照度由下式给出。

Therefore the irradiance on the image plane is

$$Irradiance = \frac{d\psi}{dA_p} = \left(\frac{\pi}{4}\right) L_{object} \left(\frac{d}{f}\right)^2 \cos^4\theta$$

The irradiance is converted to pixel intensities, which is directly proportional to the radiance of the object.

From light rays to pixel values 光线到像素值的转换



- Camera response function: the mapping f from irradiance to pixel values
  - Useful if we want to estimate material properties
  - Enables us to create *high dynamic range (HDR) images*
  - Classic reference: P. E. Debevec and J. Malik, *Recovering High Dynamic Range Radiance Maps from Photographs*, *SIGGRAPH 97*

相机响应函数表示从辐照度到像素值的映射 f。该函数有助于估计材质属性，并生成高动态范围图像（HDR）。

经典文献 P. E. Debevec 与 J. Malik《从照片恢复高动态范围辐亮度图》SIGGRAPH 1997。

Lens 镜头 Shutter 快门 CCD 电荷耦合器件 ADC 模数转换 Remapping 重映射。L 场景辐亮度 E 传感器辐照度 X 传感器曝光量 Z 最终数字值。analog voltages 模拟电压 digital values 数字值。

Photometry 光度学

Radiometry：energy，objective

Photometry：visible intensity，subjective

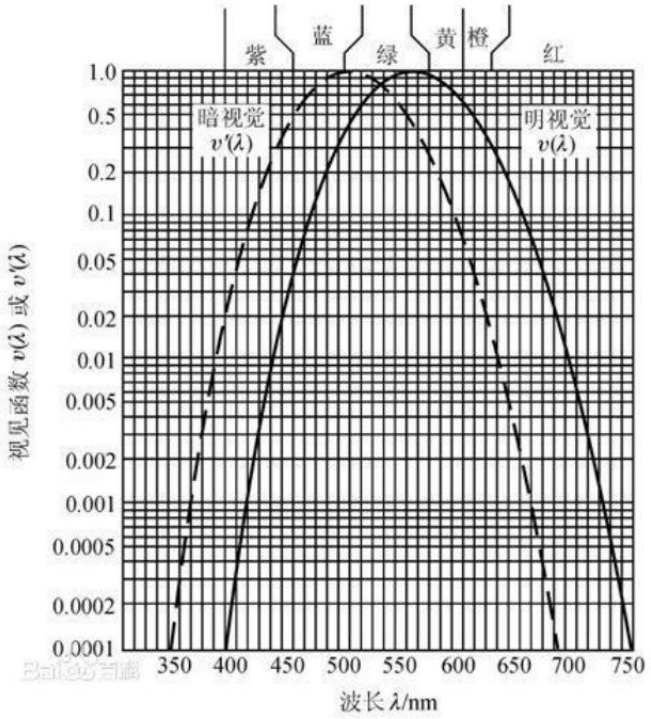
Visibility function V_λ

$\phi_\lambda = V_\lambda . P$

Definition：visible energy=683lm @λ=555nm，P=1W

$$\phi_\lambda = 683 \int_{\lambda_a}^{\lambda_b} V_\lambda \varphi_\lambda d\lambda$$

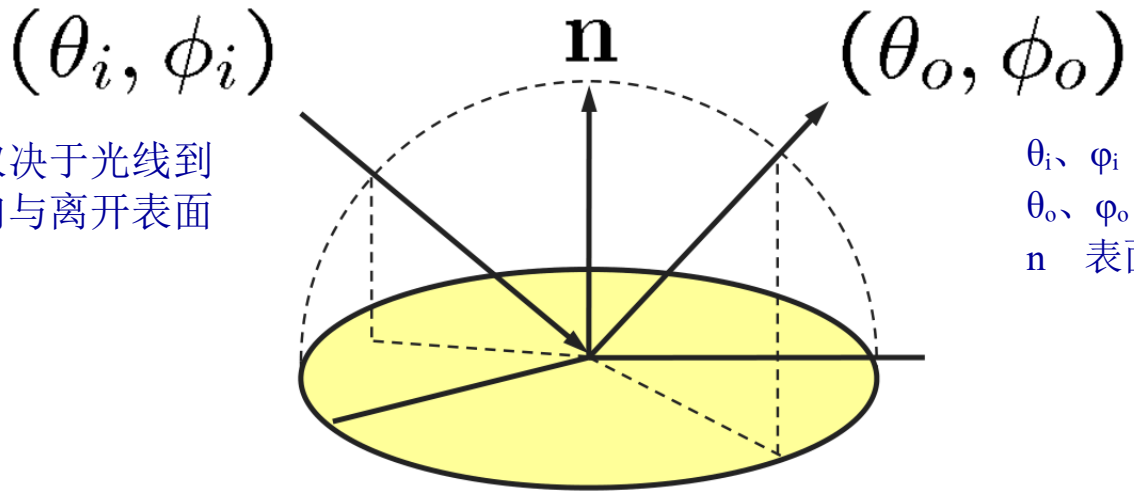
辐射度学以能量为对象，描述客观物理量。
光度学以人眼感知的可见光强弱为对象，具有主观视觉属性。
Vλ 为视见函数。
在 λ=555 nm、P=1 W 时，光通量为 683 lm。



Surface Characteristics 表面特性

- We want to describe the relationship between incoming light and reflected light.
- This is a function of both the direction in which light arrives at a surface and the direction in which it leaves.

表面特性描述入射光与反射光之间的关系。



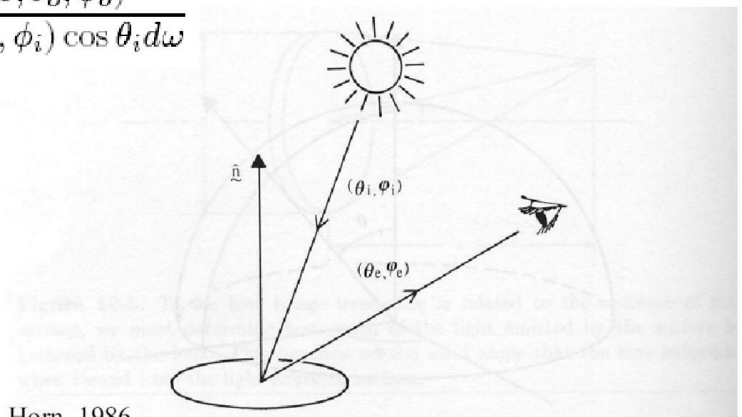
该关系同时取决于光线到达表面的方向与离开表面的方向。

θ_i、φ_i 入射方向
θ_o、φ_o 出射方向
n 表面法向

Bidirectional Reflectance Distribution Function (BRDF) 双向反射分布函数

- BRDF is defined as the ratio of the radiance in the outgoing direction to the incident irradiance.

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(x, \theta_o, \phi_o)}{L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega}$$



Horn, 1986

Figure 10-7. The bidirectional reflectance distribution function is the ratio of the radiance of the surface patch as viewed from the direction (θ_e, ϕ_e) to the irradiance resulting from illumination from the direction (θ_i, ϕ_i).

BRDF 定义为出射方向上的辐亮度与入射辐照度之比。

图 10-7 观察方向为（θ_e，φ_e）。照明方向为（θ_i，φ_i）。双向反射分布函数是沿观察方向得到的表面微元辐亮度，与照明方向所产生辐照度之比。

来源 Horn，1986。

Bidirectional Reflectance Distribution Function (BRDF) 双向反射分布函数

- The radiance leaving a surface due to irradiance *in a particular direction* is easily obtained from the definition of BRDF:

$$L_o(x, \theta_o, \phi_o) = \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega$$

依据 BRDF 的定义，可以求得特定方向的入射辐照度所产生的表面出射辐亮度。

Bidirectional Reflectance Distribution Function (BRDF) 双向反射分布函数

- The radiance leaving a surface due to irradiance *in all incoming directions* is

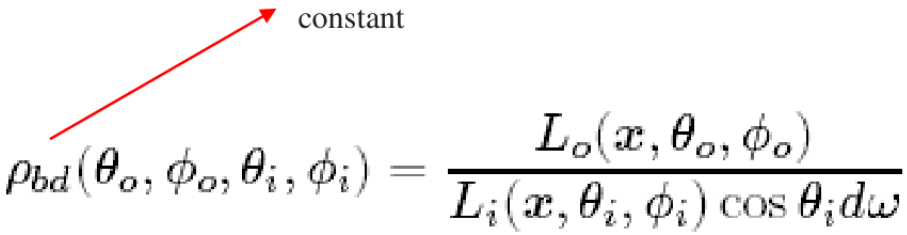
$$L_o(\boldsymbol{x}, \theta_o, \phi_o) = \int_{\Omega} \rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) L_i(\boldsymbol{x}, \theta_i, \phi_i) \cos\theta_i d\omega$$

where \Omega is the incoming hemisphere.

各入射方向的辐照度共同产生的表面出射辐亮度由式中积分给出。
Ω 表示入射半球。

Lambertian Surface　朗伯表面

- A *Lambertian surface* has constant BRDF.



朗伯表面的双向反射分布函数（BRDF）为常数。
图中 constant 表示常数。

Lambertian Surface 朗伯表面

- A Lambertian surface looks equally bright from any view direction.
- The image intensities of the surface only changes with the illumination directions.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(x, \theta_o, \phi_o)}{L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega}$$

任意观察方向下，朗伯表面呈现相同亮度。
表面的图像强度仅随光照方向变化。
图中 constant 表示常数。

Lambertian Surface 朗伯表面

- For a Lambertian surface, the outgoing radiance is proportional to the incident radiance.

constant

$$\rho_{bd}(\theta_o, \phi_o, \theta_i, \phi_i) = \frac{L_o(x, \theta_o, \phi_o)}{L_i(x, \theta_i, \phi_i) \cos\theta_i d\omega}$$

- If the light source is a point source, a pixel intensity will only be a function of $cos\theta_i$

Remember, for a point source $Irradiance = \frac{d\psi}{dA_i} = I\frac{\cos\theta_i}{r^2}$

对于朗伯表面，出射辐亮度与入射辐亮度成正比。

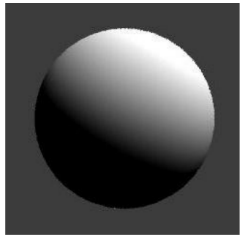
图中 constant 表示常数。

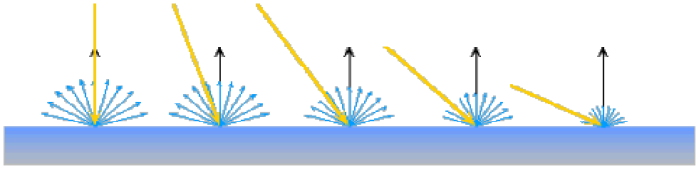
点光源照明时，像素强度仅为 $\cos\theta_i$ 的函数。

回顾点光源的辐照度关系 $E = I\cos\theta_i / r^2$。

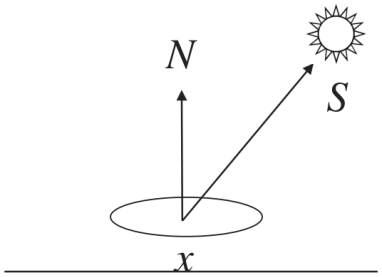
Diffuse reflection: Lambert's law 漫反射与朗伯定律

- Viewed brightness does not depend on viewing direction, but it *does* depend on direction of illumination





$$B(x) = \rho_d(x)\left(N(x)\cdot S_d(x)\right)$$



B: radiosity
ρ: albedo
N: unit normal
S: source vector (magnitude proportional to intensity of the source)

观察亮度与视线方向无关，由光照方向决定。

B 表示辐射出射度。

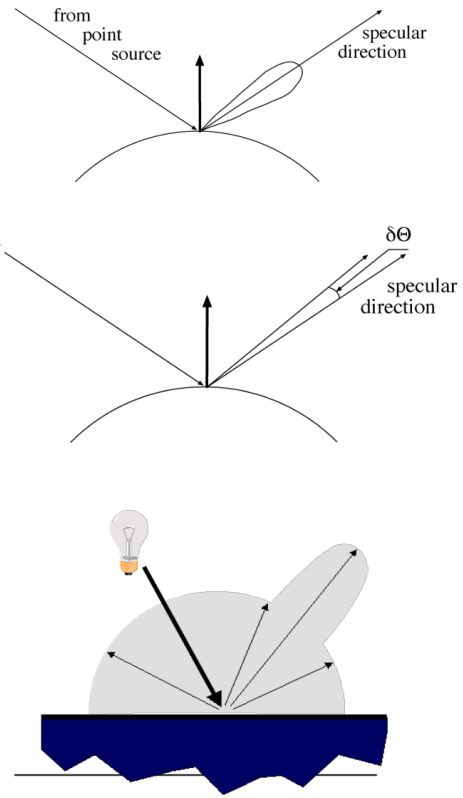
ρ 表示反照率。

N 表示单位法线。

S 表示光源向量，其模长与光源强度成正比。

Specular reflection 镜面反射

- Radiation arriving along a source direction leaves along the specular direction (source direction reflected about normal)
- Some fraction is absorbed, some reflected
- On real surfaces, energy usually goes into a lobe of directions
- Phong model: reflected energy falls of with $\cos^n(\delta\theta)$
- Lambertian + specular model: sum of diffuse and specular term



沿光源方向入射的辐射沿镜面反射方向离开。该方向由光源方向相对于法线的对称方向确定。

一部分辐射被吸收，另一部分发生反射。

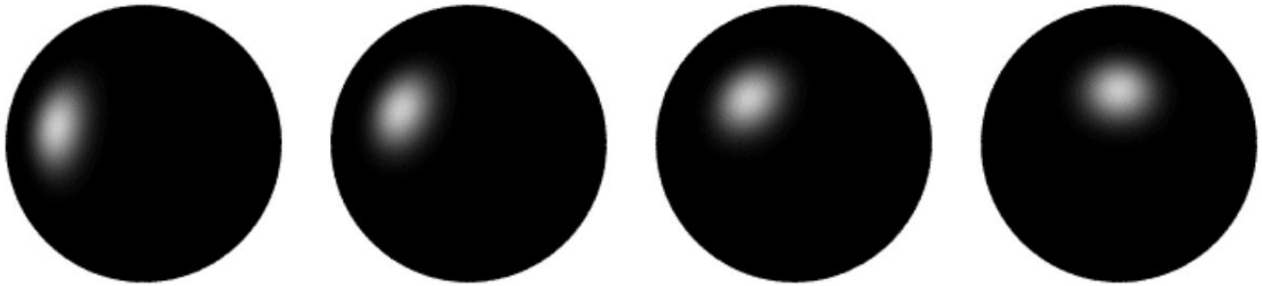
实际表面的反射能量通常分布在一个方向波瓣内。

Phong 模型中，反射能量按 $\cos^n(\delta\theta)$ 衰减。

朗伯与镜面反射组合模型等于漫反射项与镜面反射项之和。

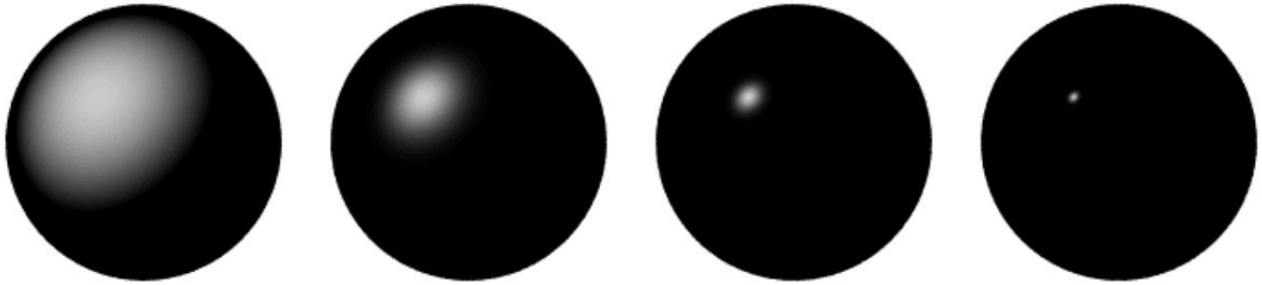
图示 from point source 表示来自点光源，specular direction 表示镜面反射方向。

Specular reflection 镜面反射



Moving the light source

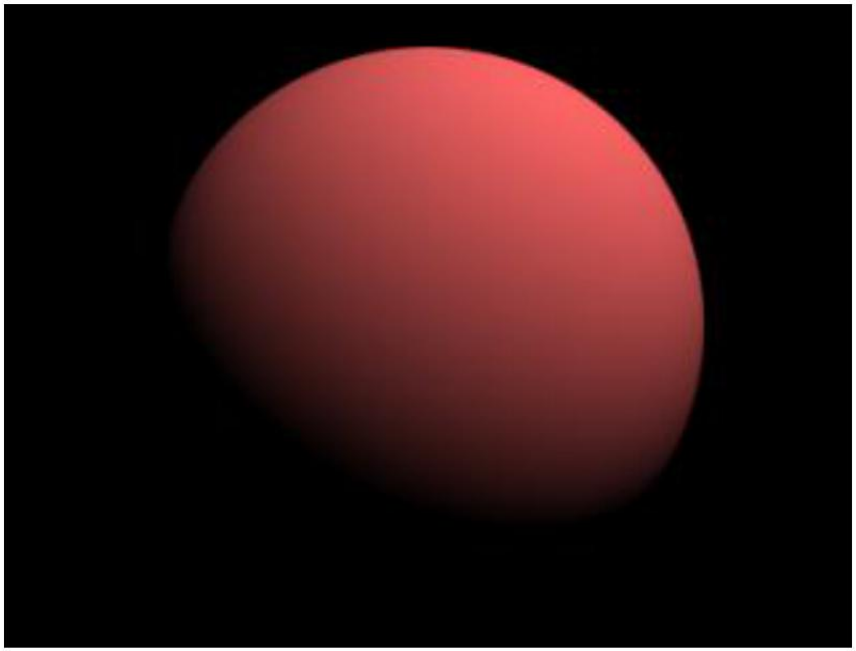
光源位置变化



Changing the exponent

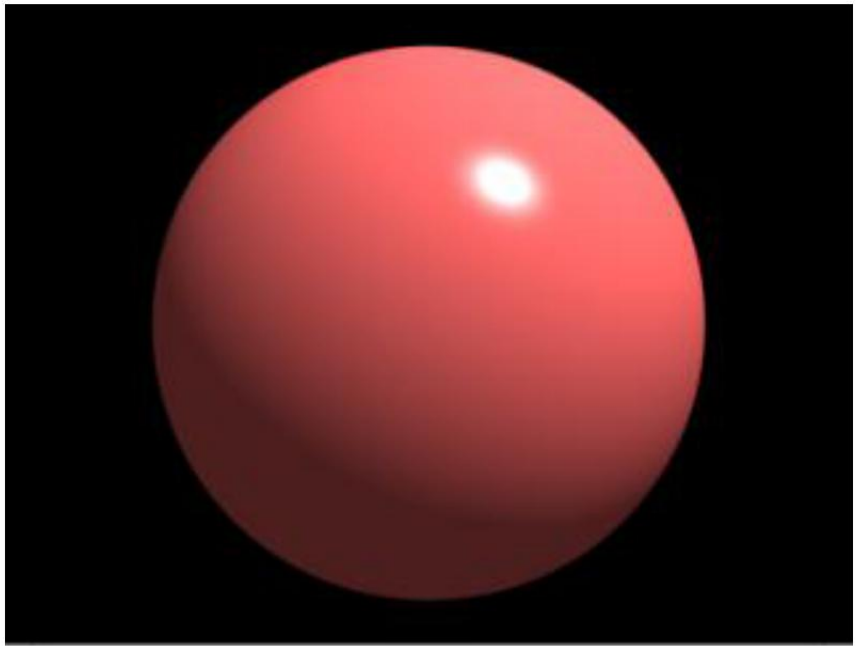
指数变化

Lambertian + Specular Model　朗伯与镜面反射组合模型



Lambertian

朗伯反射



Lambertian + Specular

朗伯与镜面反射组合

What differences in intensity tell us about shape 强度差异与形状线索

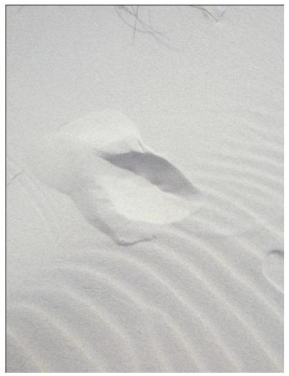
Changes in surface normal

Texture

Proximity

Indents and bumps

Grooves and creases



表面法线变化。

纹理。

邻近关系。

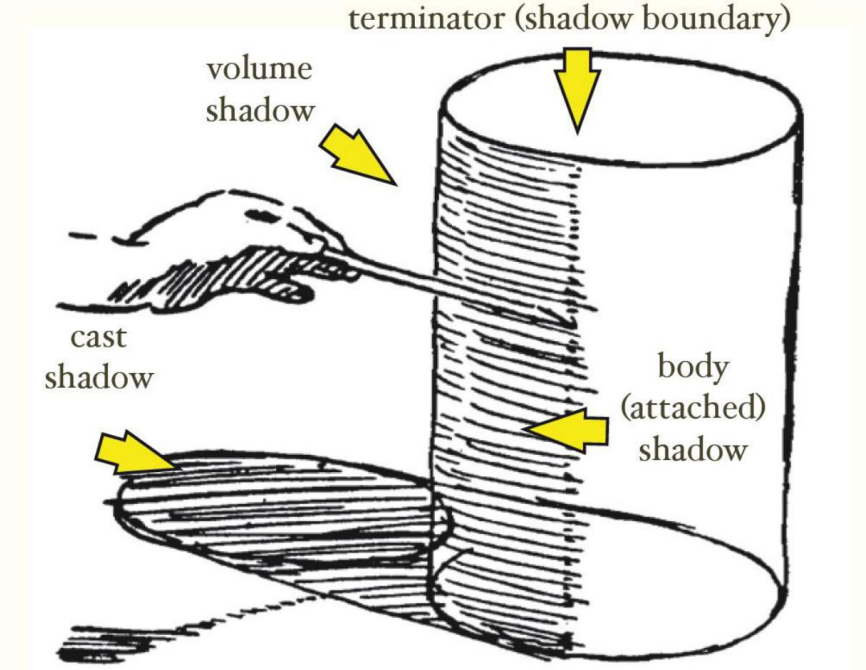
凹陷与凸起。

沟槽与褶皱。

图片源自 Koenderink 关于图像纹理与光流动的课件。

Photos Koenderink slides on image texture and the flow of light

Shadows as cues 阴影线索



From Koenderink slides on image texture and the flow of light

terminator（shadow boundary）表示明暗分界线，也称阴影边界。

volume shadow 表示体阴影。

cast shadow 表示投射阴影。

body（attached）shadow 表示物体自身的附着阴影。

来源 Koenderink 关于图像纹理与光流动的课件。

Photometric stereo (shape from shading) 光度立体与明暗恢复形状

- Can we reconstruct the shape of an object based on shading cues?

研究问题为利用明暗线索重建物体形状。

卢卡德拉罗比亚
《唱诗班席》
1438 年

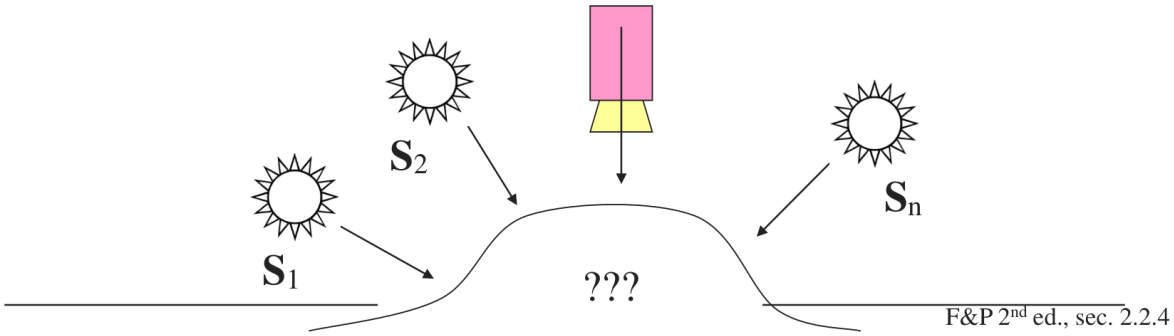
Luca della Robbia,
Cantoria, 1438

Photometric stereo 光度立体

Assume:

- A Lambertian object
- A *local shading model* (each point on a surface receives light only from sources visible at that point)
- A set of *known* light source directions
- A set of pictures of an object, obtained in exactly the same camera/object configuration but using different sources
- Orthographic projection

Goal: reconstruct object shape and albedo



F&P 2nd ed., sec. 2.2.4

模型假设如下。

物体为朗伯物体。

采用局部明暗模型，表面各点仅受该点可见光源的照明。

一组光源方向已知。

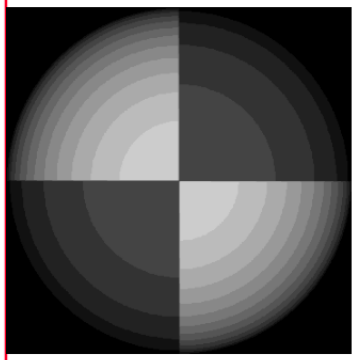
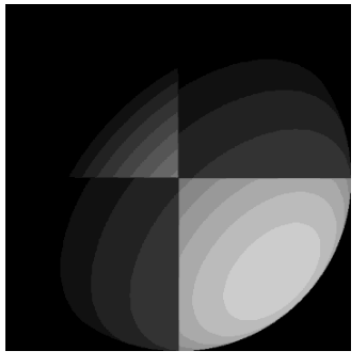
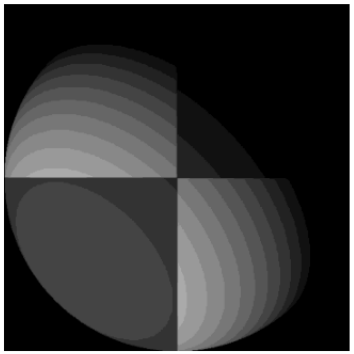
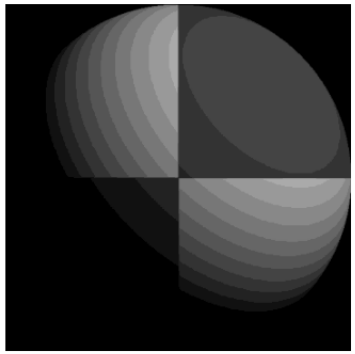
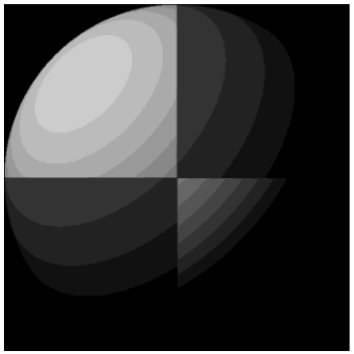
相机与物体的位置关系保持相同，使用多组光源拍摄一组图像。

采用正交投影。

目标为重建物体形状与反照率。

参考 F&P 第 2 版第 2.2.4 节。

Example 1　示例一

恢复的反照率

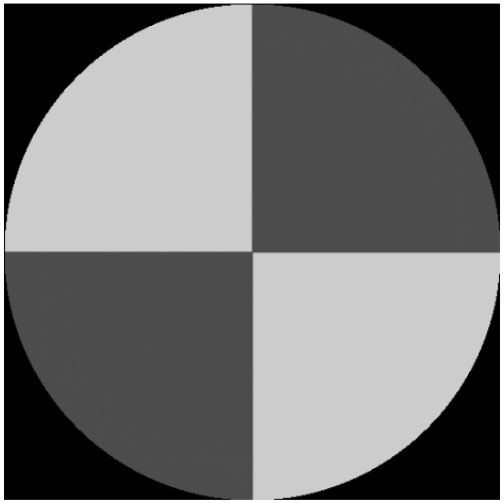
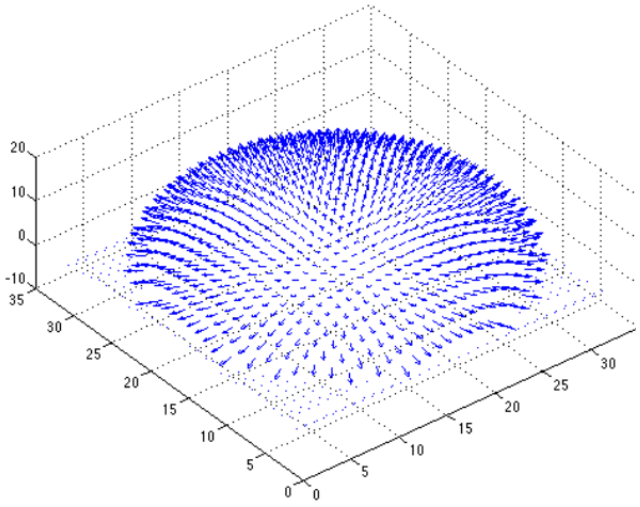
Recovered albedo

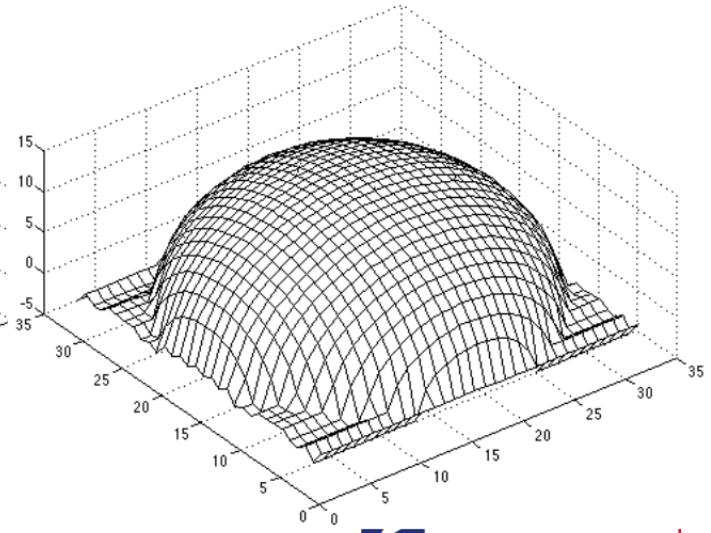
恢复的法线场

Recovered normal field

恢复的曲面模型

Recovered surface model



Example 2　示例二

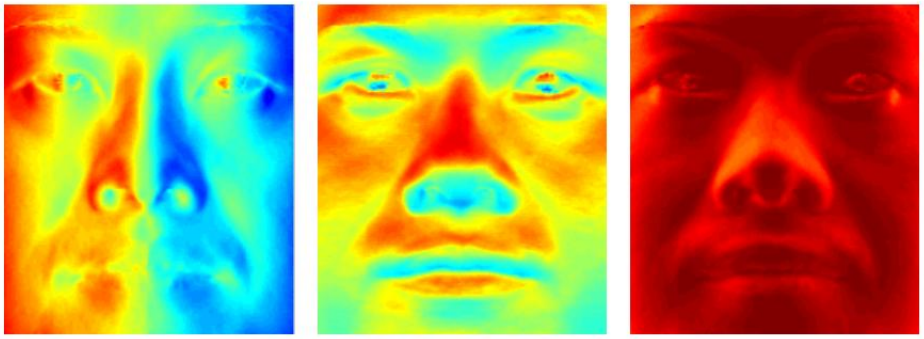
Input　输入图像

恢复的反照率

Recovered albedo

恢复的法线场

Recovered normal field



x　　y　　z

恢复的曲面模型

Recovered surface model

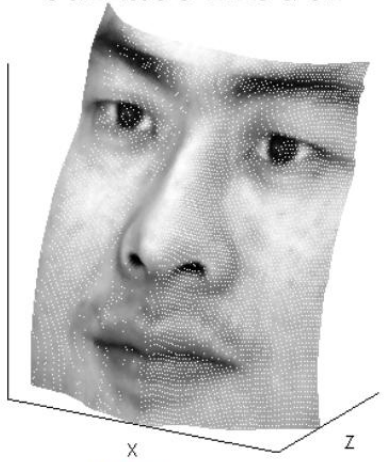
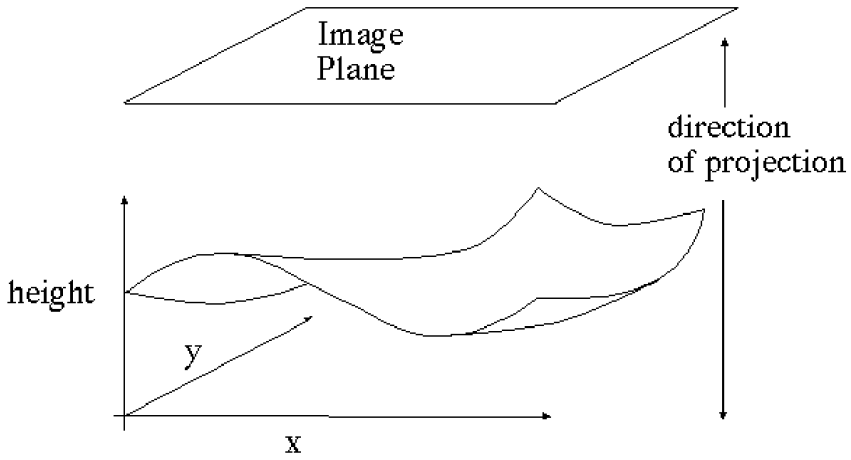


Image model　图像模型

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$



已知光源向量 S_j 与像素值 $I_j(x,y)$。

待求表面法线 $N(x,y)$ 与反照率 $\rho(x,y)$。

图中 Image Plane 表示像平面，direction of projection 表示投影方向，height 表示高度。

F&P 2nd ed., sec. 2.2.4

Image model　图像模型

- **Known:** source vectors $\mathbf{S}_j$ and pixel values $I_j(x,y)$
- **Unknown:** surface normal $\mathbf{N}(x,y)$ and albedo $\rho(x,y)$
- Assume that the response function of the camera is a linear scaling by a factor of k
- Lambert's law:

$$
\begin{aligned}
I_j(x,y) &= k\rho(x,y)\left(\mathbf{N}(x,y)\cdot\mathbf{S}_j\right) \\
&= \left(\rho(x,y)\mathbf{N}(x,y)\right)\cdot(k\mathbf{S}_j) \\
&= \mathbf{g}(x,y)\cdot\mathbf{V}_j
\end{aligned}
$$

已知光源向量 S_j 与像素值 $I_j(x,y)$。

待求表面法线 N(x,y) 与反照率 ρ(x,y)。

假设相机响应函数为系数 k 的线性缩放。

朗伯定律给出图中的像素强度关系。

将反照率与法线组合为 g(x,y)，将线性系数与光源向量组合为 V_j。

F&P 2nd ed., sec. 2.2.4

Least squares problem 最小二乘问题

- For each pixel, set up a linear system:

$$\begin{bmatrix} I_1(x,y) \\ I_2(x,y) \\ \vdots \\ I_n(x,y) \end{bmatrix} = \begin{bmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \\ \vdots \\ \mathbf{V}_n^T \end{bmatrix} \mathbf{g}(x,y)$$

$(n \times 1)$ known　　$(n \times 3)$ known　　(3×1) unknown

- Obtain least-squares solution for $\mathbf{g}(x,y)$ (which we defined as $\mathbf{N}(x,y)\,\rho(x,y)$)
- Since $\mathbf{N}(x,y)$ is the unit normal, $\rho(x,y)$ is given by the magnitude of $\mathbf{g}(x,y)$
- Finally, $\mathbf{N}(x,y) = \mathbf{g}(x,y) / \rho(x,y)$

F&P 2nd ed., sec. 2.2.4

为每个像素建立一个线性方程组。

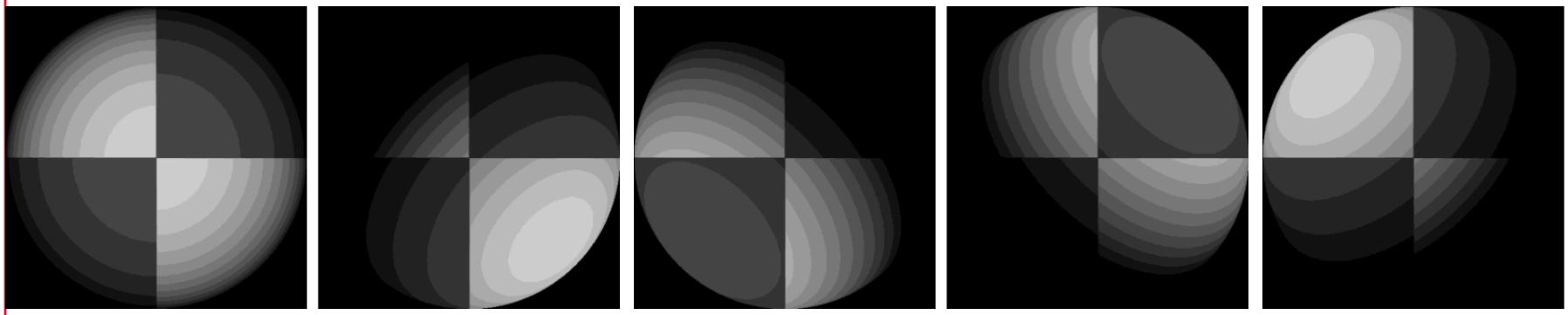
图中 n × 1 的强度向量和 n × 3 的光源矩阵已知，3 × 1 的 g(x,y) 向量待求。

求解 g(x,y) 的最小二乘解，其中 g(x,y) 定义为 N(x,y)ρ(x,y)。

N(x,y) 为单位法线，因此 ρ(x,y) 等于 g(x,y) 的模长。

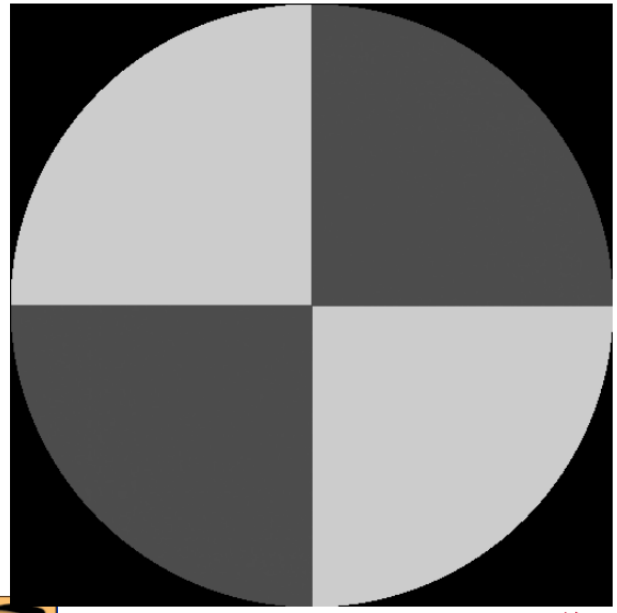
由 N(x,y) = g(x,y) /ρ(x,y) 恢复法线。

Synthetic example 合成示例



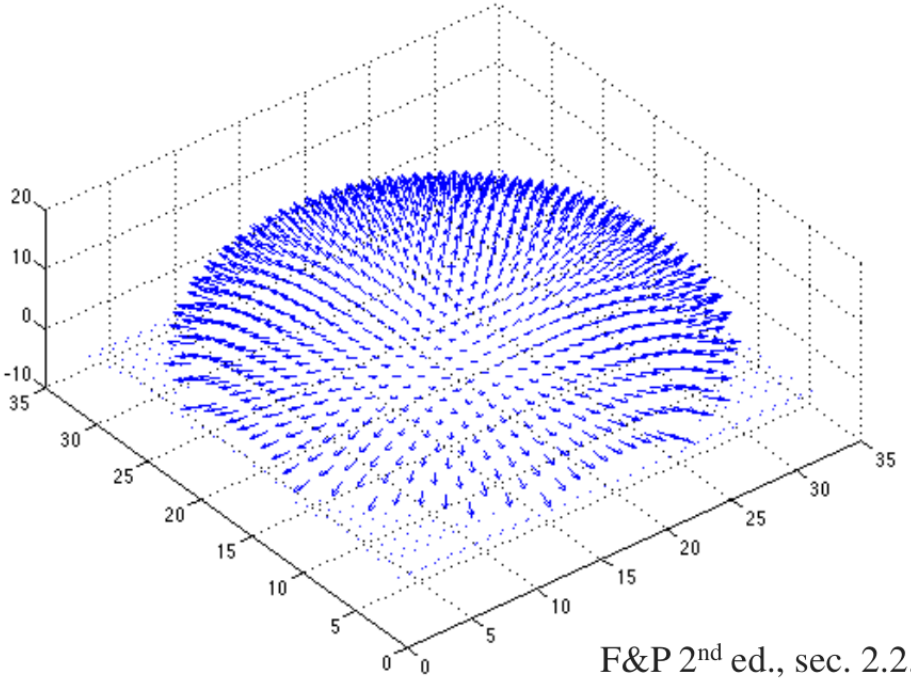
恢复的反照率

Recovered albedo



恢复的法线场

Recovered normal field



华中

F&P 2nd ed., sec. 2.2.4

Recovering a surface from normals 由法线恢复曲面

Recall the surface is written as

$$(x, y, f(x, y))$$

This means the normal has the form:

$$\mathbf{N}(x, y) = \frac{1}{\sqrt{f_x^2 + f_y^2 + 1}} \begin{pmatrix} f_x \\ f_y \\ 1 \end{pmatrix}$$

If we write the estimated vector g as

$$\mathbf{g}(x, y) = \begin{pmatrix} g_1(x, y) \\ g_2(x, y) \\ g_3(x, y) \end{pmatrix}$$

Then we obtain values for the partial derivatives of the surface:

$$f_x(x, y) = g_1(x, y) / g_3(x, y)$$
$$f_y(x, y) = g_2(x, y) / g_3(x, y)$$

F&P 2nd ed., sec. 2.2.4

曲面可表示为 (x,y, f(x,y))。

由此得到图中的单位法线形式。

将估计向量 g 写为三个分量 g_1、g_2、g_3。

由这些分量求得曲面的偏导数 f_x 与 f_y。

Recovering a surface from normals 由法线恢复曲面

We can now recover the surface height at any point by integration along some path, e.g.

$$f(x,y) = \int_0^x f_x(s,0)\,ds + \int_0^y f_y(x,t)\,dt + C$$

(for robustness, should take integrals over many different paths and average the results)

Integrability: for the surface *f* to exist, the mixed second partial derivatives must be equal:

$$\frac{\partial}{\partial y}(g_1(x,y)/g_3(x,y)) = \frac{\partial}{\partial x}(g_2(x,y)/g_3(x,y))$$

(in practice, they should at least be similar)

F&P 2nd ed., sec. 2.2.4

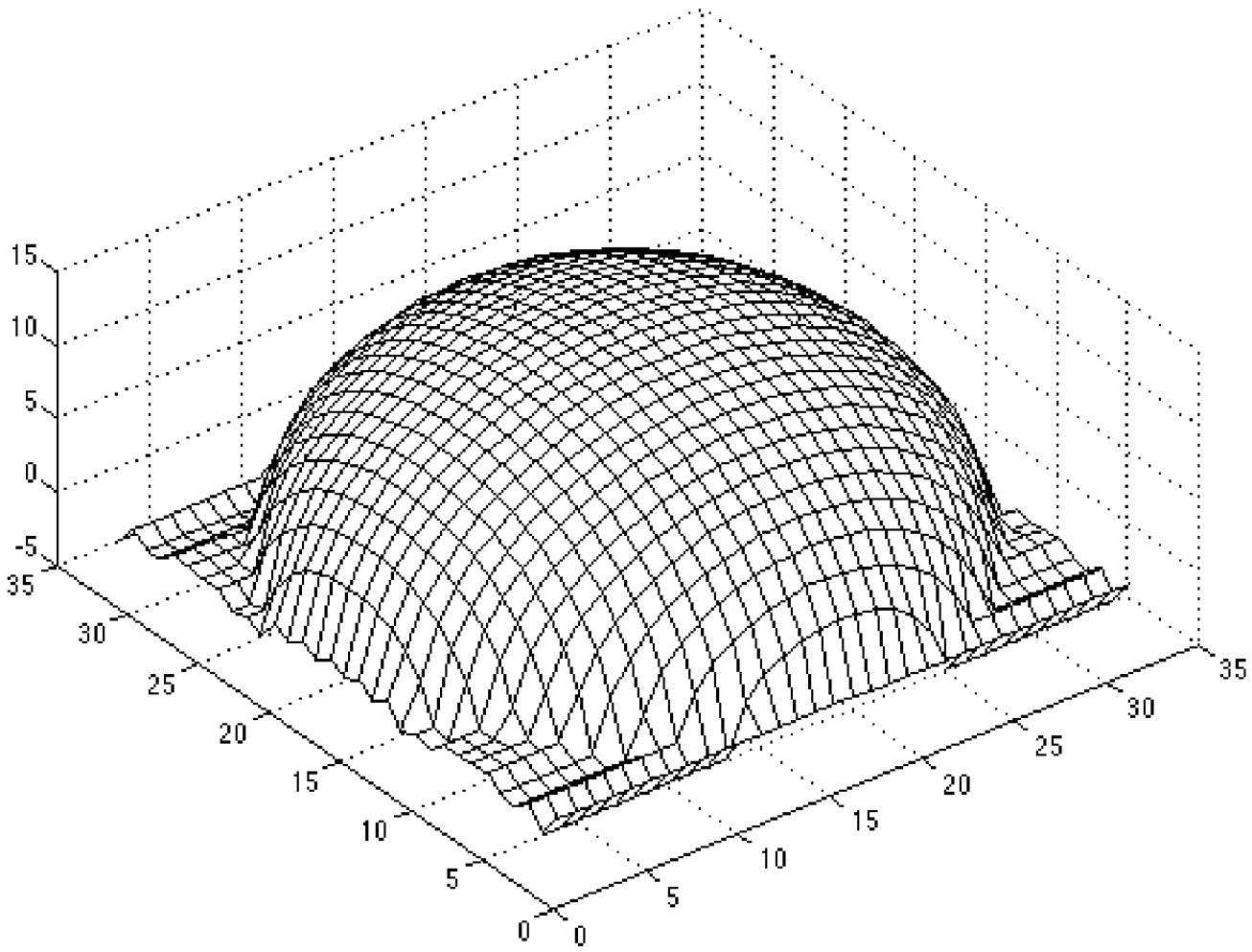
沿某条路径积分，可恢复任意点处的曲面高度，图中给出一种路径。

为减小扰动影响，可沿多条路径积分并对结果取平均。

可积性要求曲面 f 的混合二阶偏导数相等。

实际计算中，这两个偏导数至少应当相近。

Surface recovered by integration 积分恢复的曲面



Limitations 方法局限

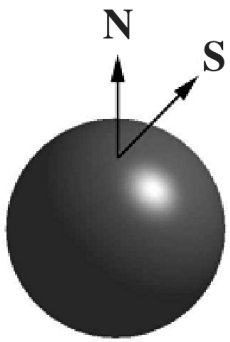
- Orthographic camera model
- Simplistic reflectance and lighting model
- No shadows
- No interreflections
- No missing data
- Integration is tricky

正交相机模型
简化的反射与光照模型
无阴影假设
忽略表面间反射
数据完整假设
积分求解较为复杂

Finding the direction of the light source　光源方向估计

$$I(x,y) = \mathbf{N}(x,y) \cdot \mathbf{S}(x,y)$$

Full 3D case:



$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

完整三维情形的光源方向由图中线性系统求解。

N 表示表面法线，S 表示光源向量。

参考文献主题为自动估计光源的投影方向，P. Nillius 与 J.-O. Eklundh，CVPR 2001。

P. Nillius and J.-O. Eklundh, "Automatic estimation of the projected light source direction," CVPR 2001

Finding the direction of the light source 光源方向估计

Consider points on the *occluding contour*:

考察遮挡轮廓上的点。

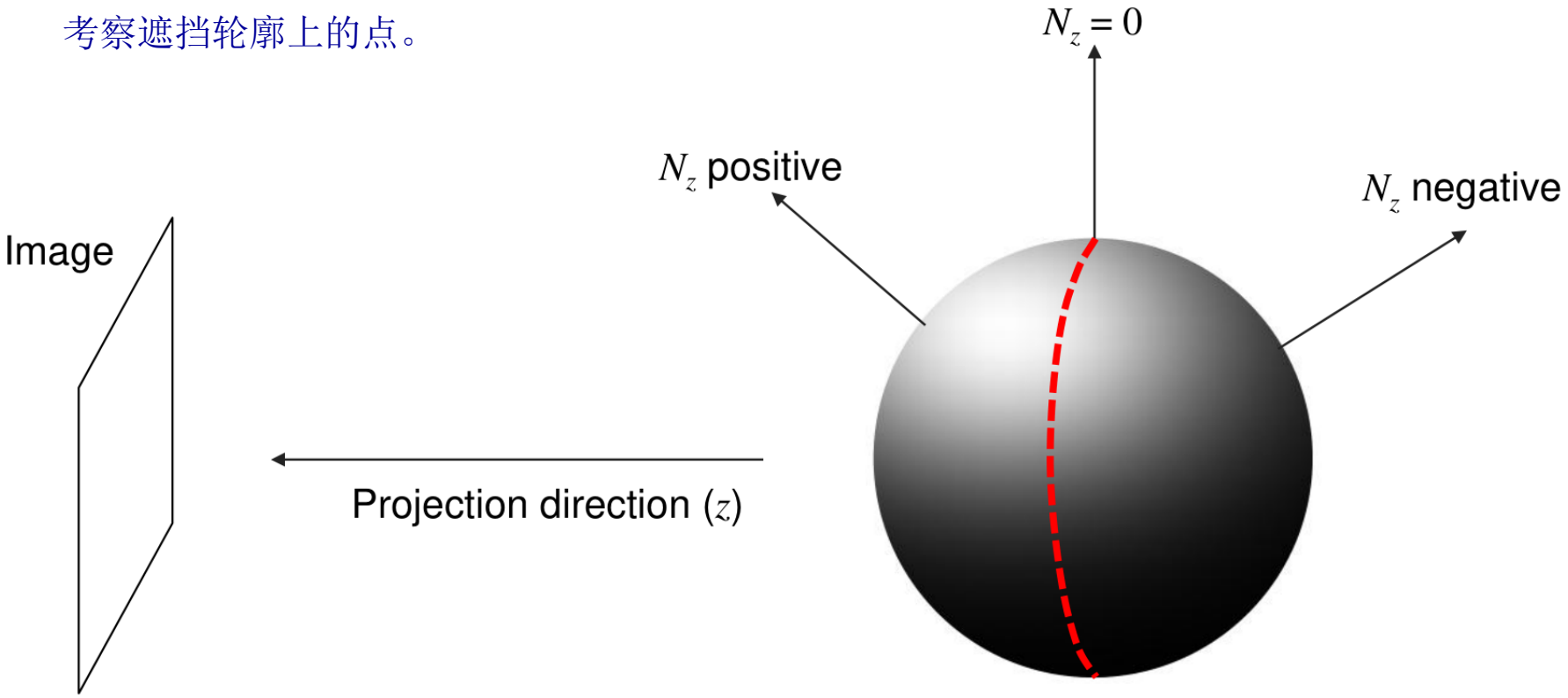
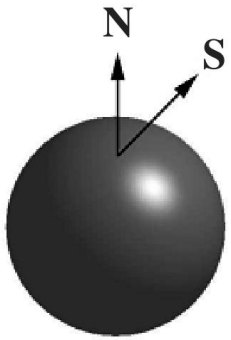


Image 为图像，Projection direction (z) 为投影方向。
N_z positive 与 N_z negative 分别表示 N_z 为正、为负。
参考文献主题为光源投影方向的自动估计（CVPR 2001）。

P. Nillius and J.-O. Eklundh, "Automatic estimation of the projected light source direction,"

Finding the direction of the light source 光源方向估计

$$I(x,y) = \mathbf{N}(x,y) \cdot \mathbf{S}(x,y)$$



Full 3D case:

$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) & N_z(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) & N_z(x_2,y_2) \\ \vdots & \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) & N_z(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \\ S_z \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

For points on the *occluding contour*, $N_z = 0$:

$$\begin{pmatrix} N_x(x_1,y_1) & N_y(x_1,y_1) \\ N_x(x_2,y_2) & N_y(x_2,y_2) \\ \vdots & \vdots \\ N_x(x_n,y_n) & N_y(x_n,y_n) \end{pmatrix} \begin{pmatrix} S_x \\ S_y \end{pmatrix} = \begin{pmatrix} I(x_1,y_1) \\ I(x_2,y_2) \\ \vdots \\ I(x_n,y_n) \end{pmatrix}$$

P. Nillius and J.-O. Eklundh, "Automatic estimation of the projected light source direction," CVPR 2001

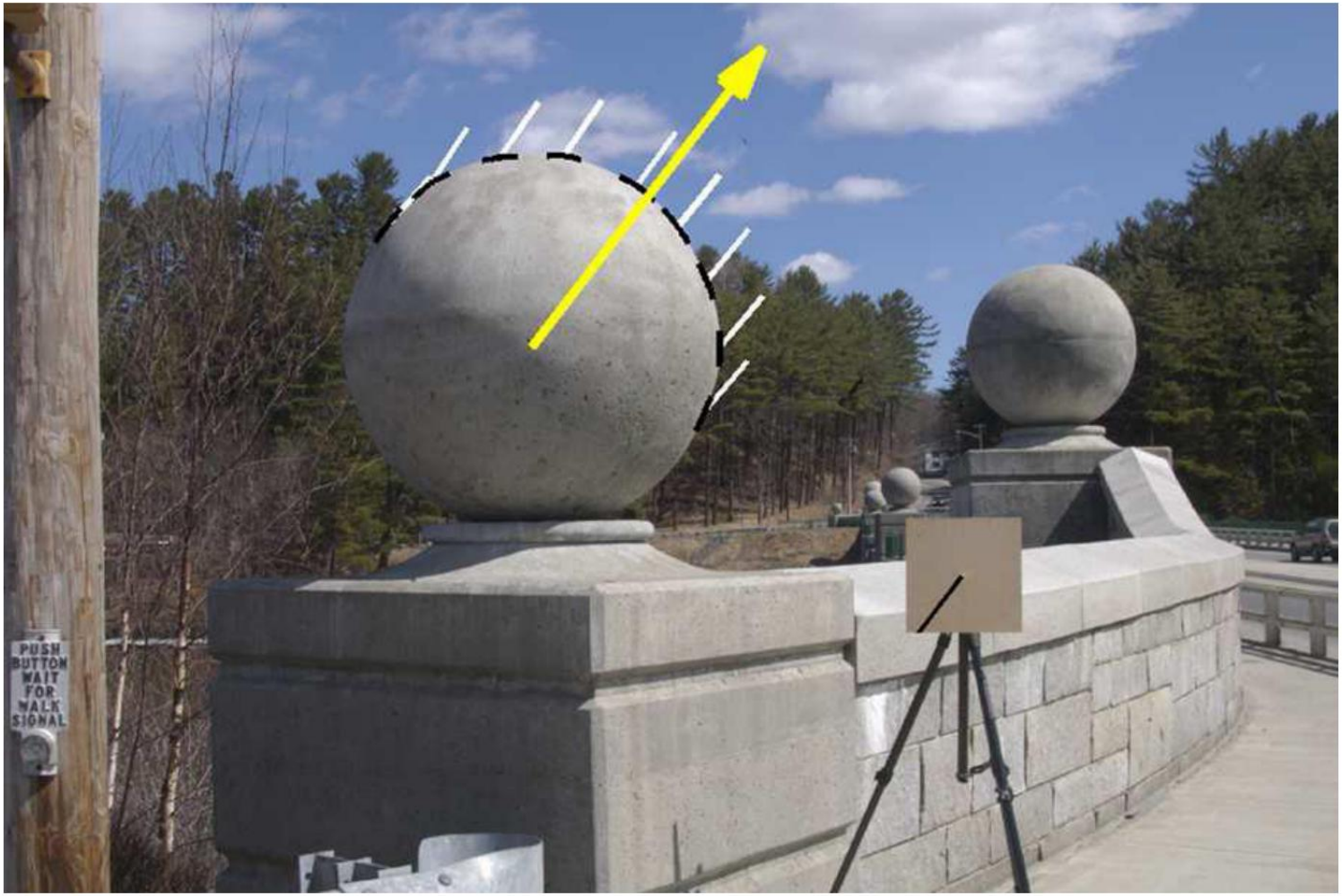
完整三维情形使用法线的三个分量 x、y、z，求解光源向量。

对于遮挡轮廓上的点，N_z = 0，可简化为图中的二维线性系统。

N 表示表面法线，S 表示光源向量。

参考文献主题为光源投影方向的自动估计，P. Nillius 与 J.-O. Eklundh，CVPR 2001。

Finding the direction of the light source 光源方向估计



参考文献主题为光源投影方向的自动估计（CVPR 2001）。

P. Nillius and J.-O. Eklundh, “Automatic estimation of the projected light source direction,”

Application: Detecting composite photos　合成照片检测应用

Real photo

Fake photo

Fake photo 表示合成照片。

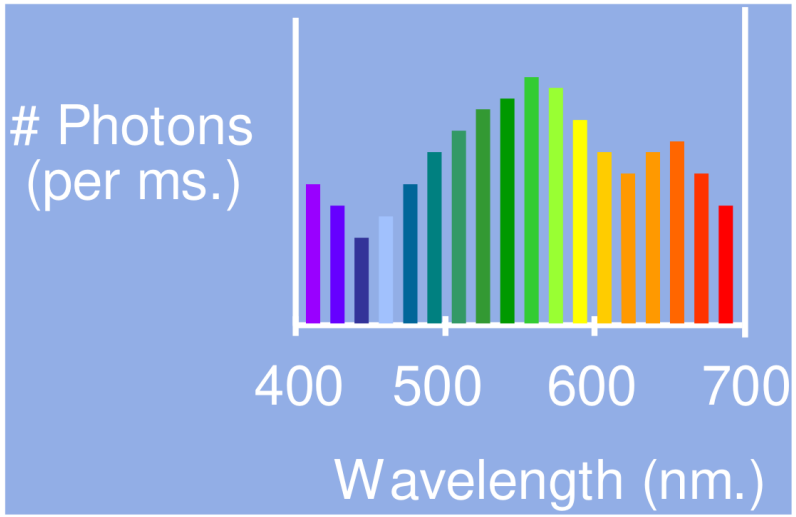
Real photo 表示真实照片。

参考文献主题为利用光照差异揭示数字图像伪造，M. K. Johnson 与 H. Farid，ACM 多媒体与安全研讨会，2005 年。

M. K. Johnson and H. Farid, Exposing Digital Forgeries by Detecting Inconsistencies in Lighting, ACM Multimedia and Security Workshop, 2005.

The Physics of Light 光的物理性质

Any patch of light can be completely described physically by its spectrum: the number of photons (per time unit) at each wavelength 400 - 700 nm.

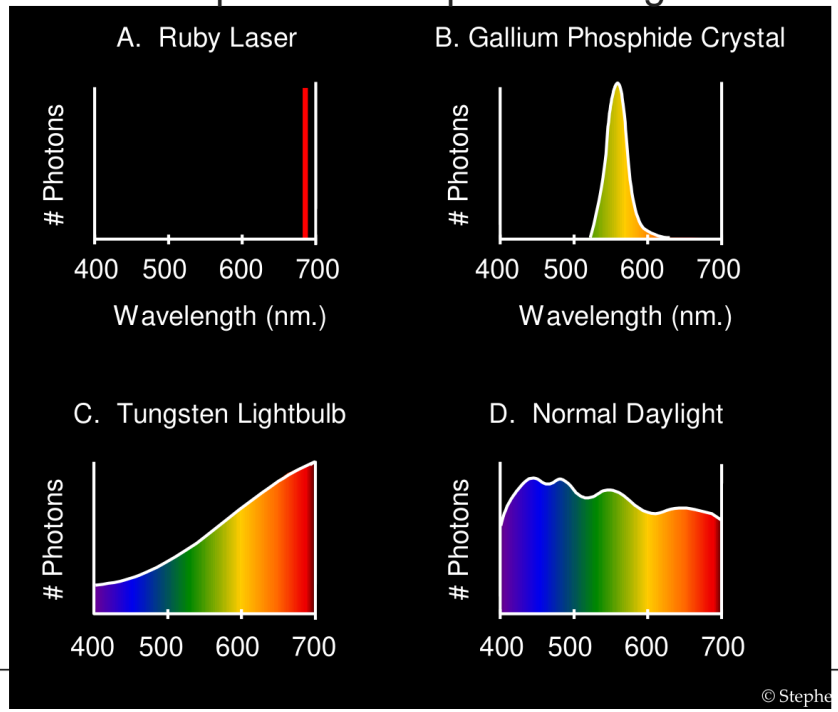


任意一束光都可用其光谱作完整的物理描述，即 400—700 nm 范围内，各波长处单位时间的光子数。

图中纵轴为每毫秒的光子数，横轴为波长，单位为 nm。

The Physics of Light 光的物理性质

Some examples of the spectra of light sources



本页展示几种光源的光谱。

A 为红宝石激光器。

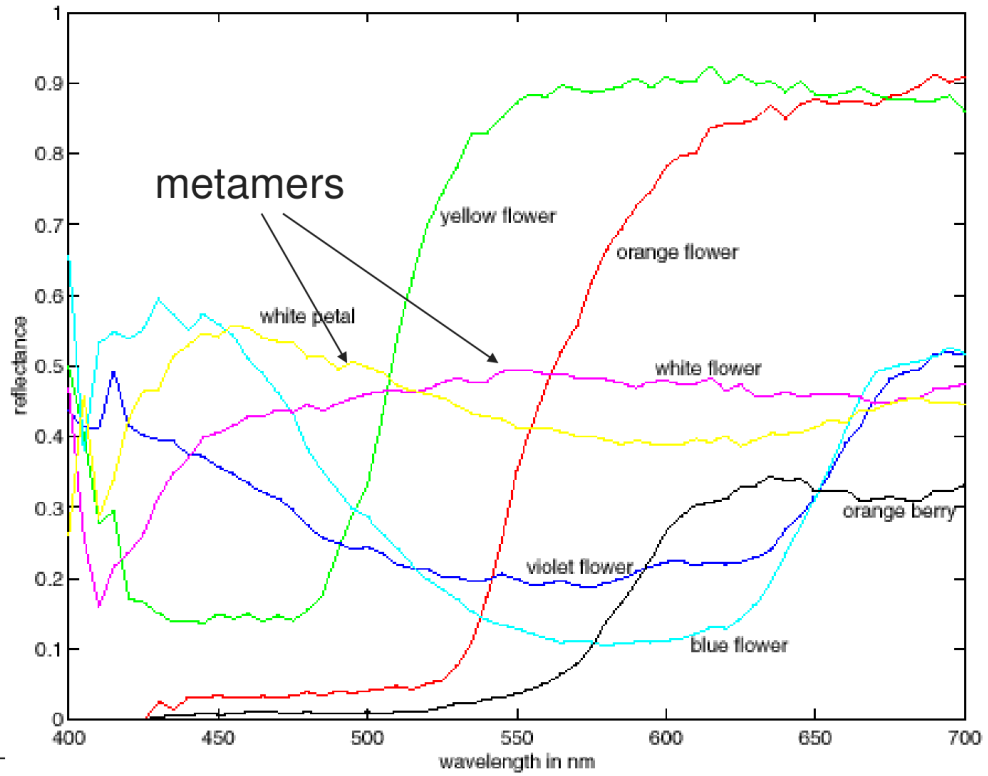
B 为磷化镓晶体。

C 为钨丝灯泡。

D 为普通日光。

各图纵轴为光子数，横轴为波长，单位为 nm。

More Spectra　更多光谱示例



metamers 表示同色异谱。

纵轴 reflectance 表示反射率，横轴表示波长，单位为 nm。

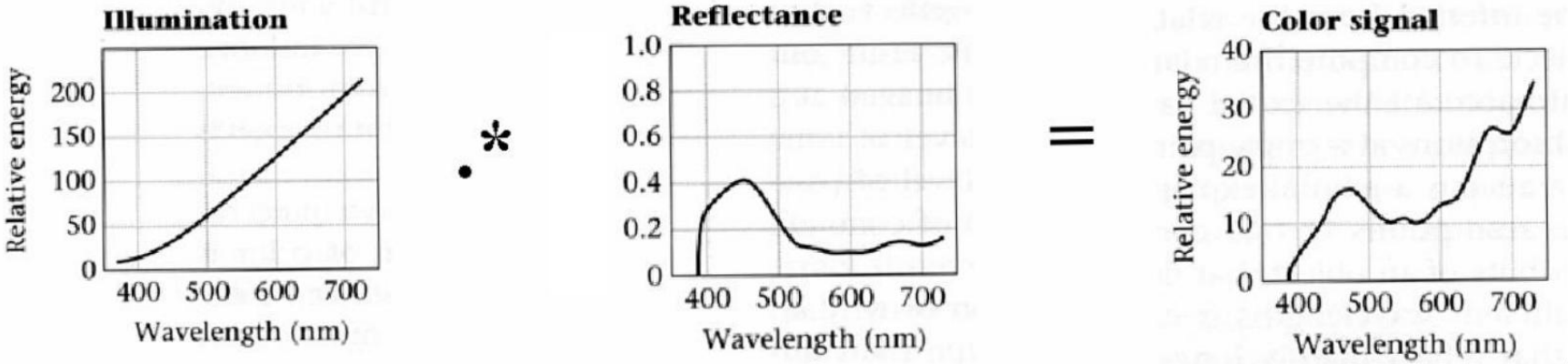
曲线标签依次涉及白色花瓣、黄色花、橙色花、白色花、橙色浆果、紫色花和蓝色花。

Interaction of light and surfaces　光与表面的相互作用

Reflected color is the result of interaction of light source spectrum with surface reflectance

反射颜色由光源光谱与表面反射率共同作用产生。图中横轴为波长（nm），纵轴为相对能量。

光照 × 反射率 = 颜色信号



Interaction of light and surfaces 光与表面的相互作用

What is the observed color of any surface under monochromatic light?

Olafur Eliasson, *Room for one color*

讨论单色光照明下任意表面的观测颜色。

作品为 Olafur Eliasson 的《单色房间》。

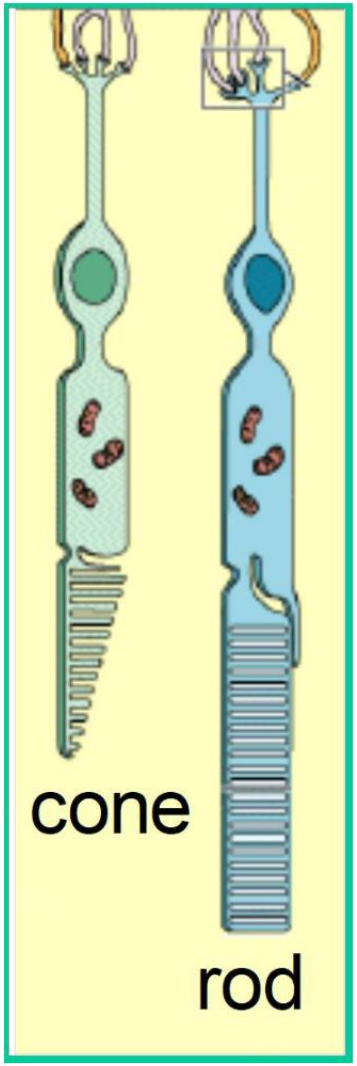
Two types of light-sensitive receptors 两类感光细胞

Cones

cone-shaped
less sensitive
operate in high light
color vision

Rods

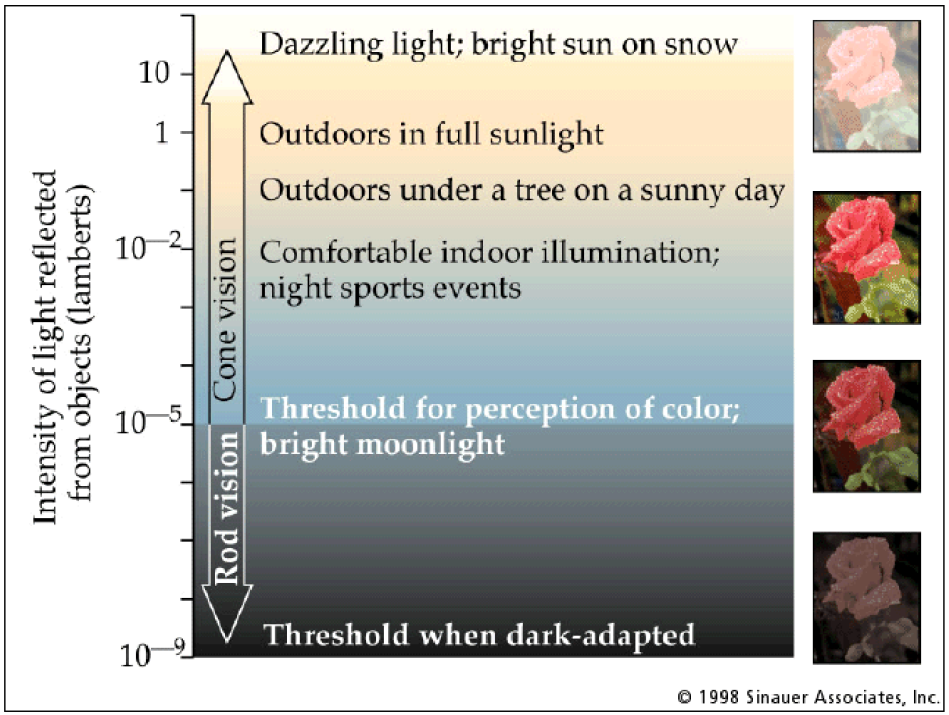
rod-shaped
highly sensitive
operate at night
gray-scale vision



视锥细胞
锥形
灵敏度较低
强光下工作
彩色视觉

视杆细胞
杆形
灵敏度很高
夜间工作
灰度视觉

Rod / Cone sensitivity　视杆与视锥细胞灵敏度



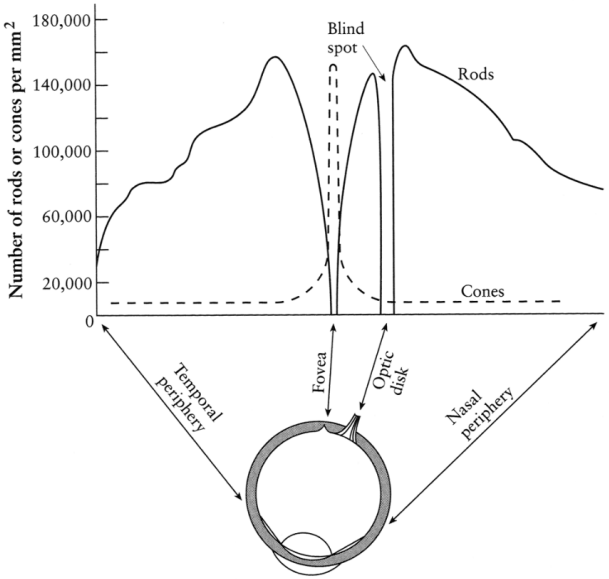
纵轴为物体反射光的亮度，单位为朗伯（lambert）。

亮度由高至低依次为耀眼强光与雪地阳光、充分日照的室外、晴天树荫下、舒适的室内照明与夜间体育赛事、色觉阈值与明亮月光、暗适应后的感知阈值。

Cone vision 为视锥视觉，Rod vision 为视杆视觉。

图中视锥视觉覆盖较高亮度范围，视杆视觉延伸至较低亮度范围。

Distribution of Rods and Cones　视杆与视锥细胞分布



Night Sky: why are there more stars off-center?
Averted vision: http://en.wikipedia.org/wiki/Averted_vision

纵轴为每平方毫米的视杆或视锥细胞数量。

Rods 为视杆细胞，Cones 为视锥细胞。

Blind spot 为盲点，Fovea 为中央凹，Optic disk 为视盘。

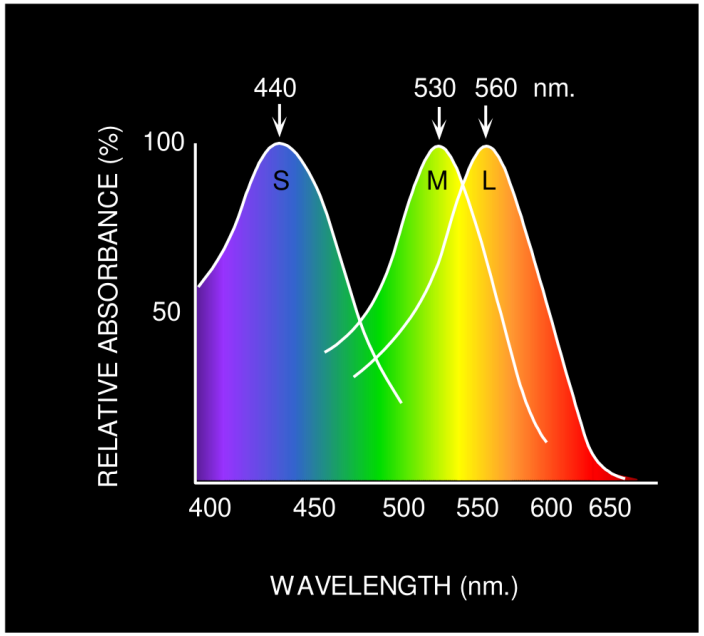
Temporal periphery 为颞侧周边，Nasal periphery 为鼻侧周边。

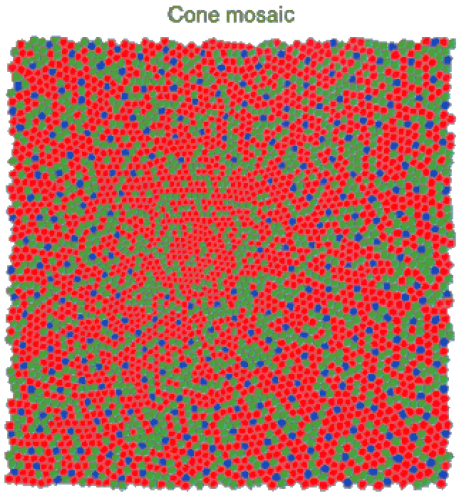
夜空观察问题为视野偏离中心时可见星点增多的原因。

Averted vision 为侧视法。

Physiology of Color Vision 颜色视觉的生理基础

Three kinds of cones:





- Why are M and L cones so close?
- Why are there 3?

三类视锥细胞分别为短波 S、中波 M、长波 L。

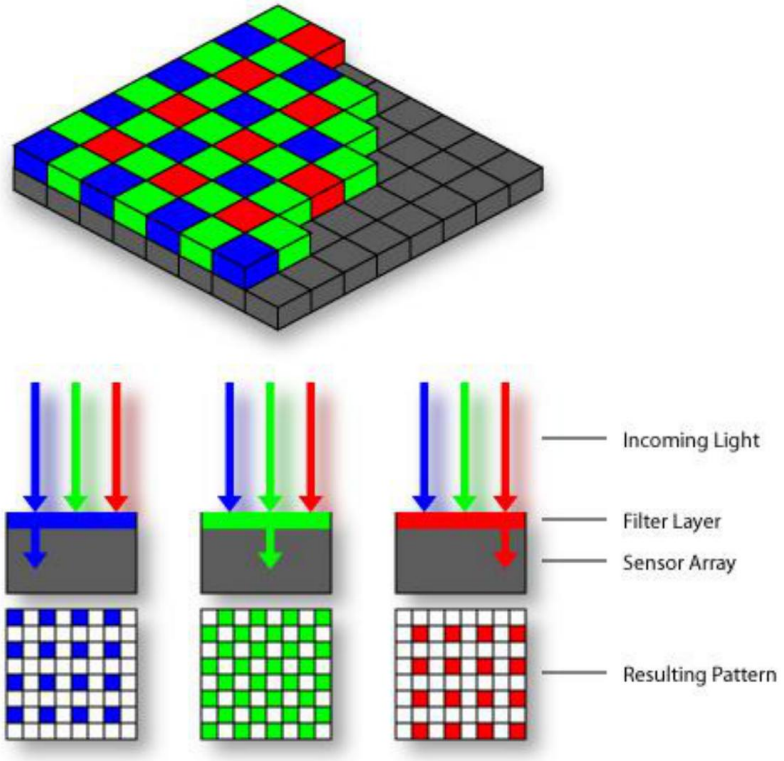
图中响应峰值约为 440、530、560 nm。

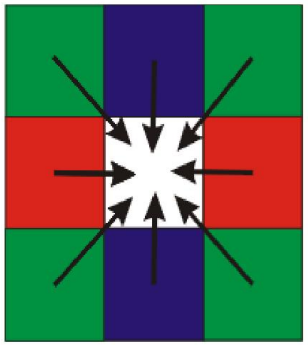
纵轴为相对吸光度，单位为 %。横轴为波长，单位为 nm。

Cone mosaic 为视锥细胞镶嵌分布。

讨论问题包括 M 与 L 视锥细胞的响应曲线为何相邻，以及视锥类型为何有三种。

Practical Color Sensing: Bayer Grid　颜色采样与拜耳阵列





Estimate RGB at 'G' cells from neighboring values

Slide by Steve Seitz

根据相邻像素值，估计 G 单元处的 RGB 值。

Incoming Light 为入射光。

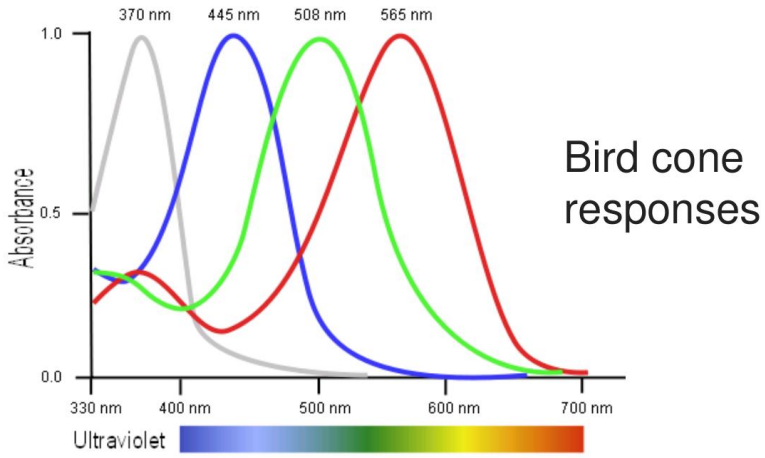
Filter Layer 为滤色层。

Sensor Array 为传感器阵列。

Resulting Pattern 为形成的采样图案。

课件图示由 Steve Seitz 提供。

Tetrachromatism 四色视觉



Bird cone responses

Most birds, and many other animals, have cones for ultraviolet light.

Some humans, mostly female, seem to have slight tetrachromatism.

多数鸟类与许多其他动物具有对紫外光敏感的视锥细胞。

部分人类似乎具有轻微的四色视觉，这类情况以女性为主。

Bird cone responses 为鸟类视锥细胞响应。

Absorbance 为吸光度，Ultraviolet 为紫外光。

Physiology of Color Vision: Fun facts 颜色视觉的生理学知识

"M" and "L" pigments are encoded on the X-chromosome

- That's why men are more likely to be color blind
- "L" gene has high variation, so some women may be *tetrachromatic*

Some animals have one (night animals), two (e.g., dogs), four (fish, birds), five (pigeons, some reptiles/amphibians), or even 12 (mantis shrimp) types of cones

http://ngm.nationalgeographic.com/2016/02/evolution-of-eyes-text

http://en.wikipedia.org/wiki/Color_vision

Slide by D. Hoiem

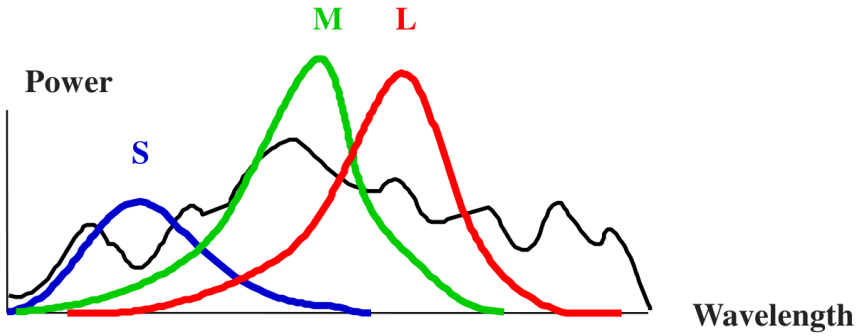
编码 M 与 L 感光色素的基因位于 X 染色体上，因此男性出现色盲的概率更高。

L 基因的变异程度较高，部分女性可能具有四色视觉。

动物视锥类型数量为夜行动物一种、犬类两种、鱼类与鸟类四种、鸽子及部分爬行动物与两栖动物五种，螳螂虾甚至有十二种。

来源 D. Hoiem。

Color perception 颜色感知



Rods and cones act as *filters* on the spectrum

To get the output of a filter, multiply its response curve by the spectrum, integrate over all wavelengths

Each cone yields one number

- How can we represent an entire spectrum with three numbers?
- We can't! Most of the information is lost
  - As a result, two different spectra may appear indistinguishable
    - » such spectra are known as **metamers**

Slide by Steve Seitz

视杆与视锥细胞对光谱起滤波作用。

将响应曲线与光谱相乘，并对所有波长积分，即得滤波器输出。

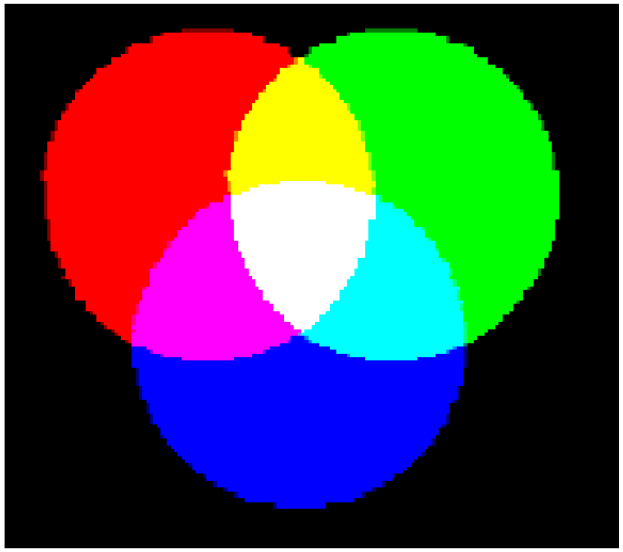
每个视锥细胞产生一个数值。

三个数值表示整段光谱时，大部分光谱信息会丢失。

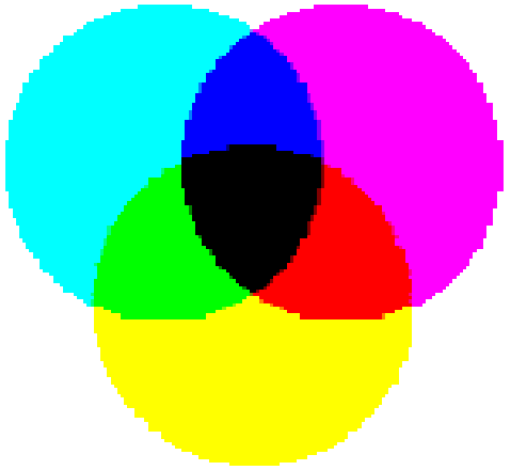
因此，两种各异的光谱可能呈现相同的感知颜色，这种现象称为同色异谱。

Power 为功率。Wavelength 为波长。S、M、L 为三类视锥响应。

颜色模型



加色混合



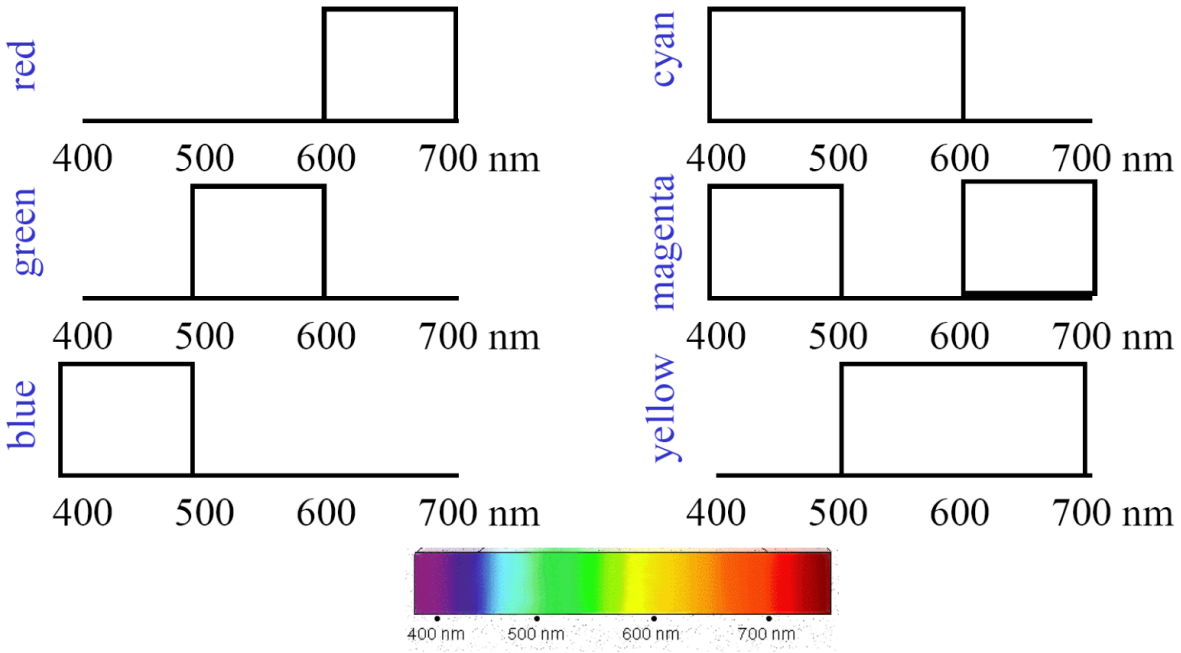
减色混合

RGB,HIS,YIQ

RGB、HSI、YIQ 颜色模型
图中 HIS 即 HSI。

Color mixing 颜色混合

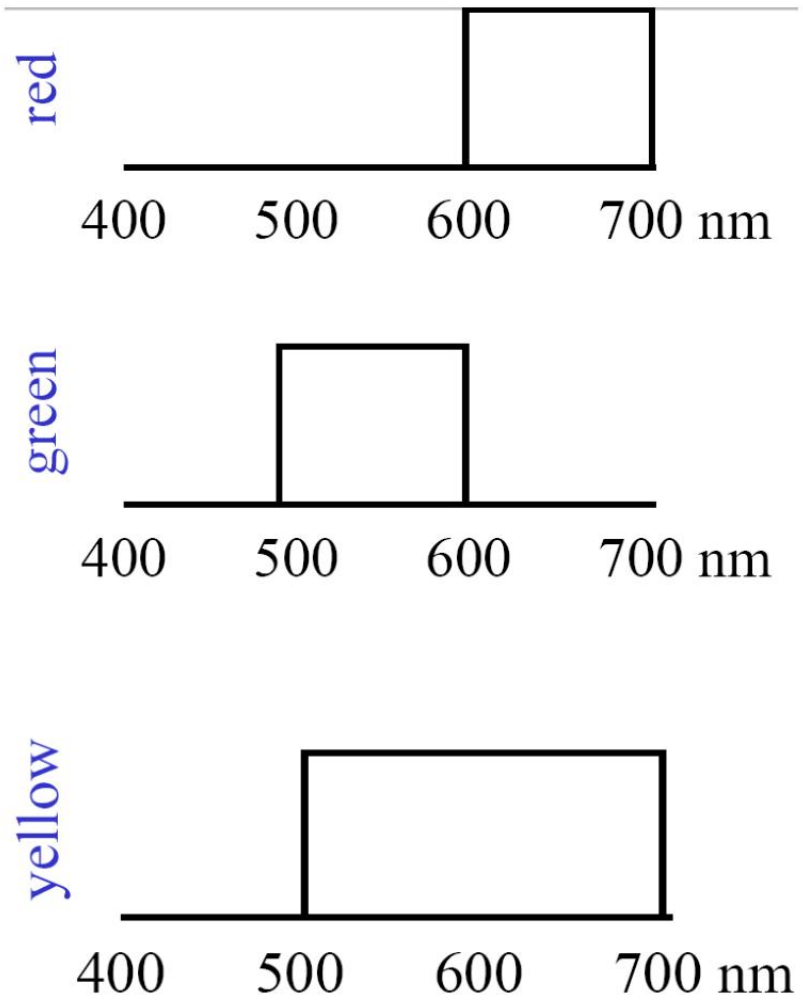
Cartoon spectra for color names:



Source: W. Freeman

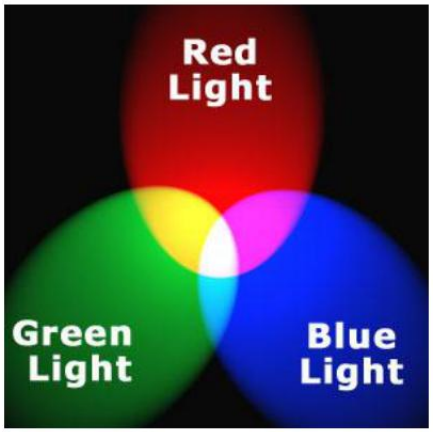
各颜色名称对应的光谱示意。red 红色，green 绿色，blue 蓝色，cyan 青色，magenta 品红色，yellow 黄色。波长单位为纳米 nm。

Additive color mixing　加色混合



各颜色的光谱相加，形成混合颜色。

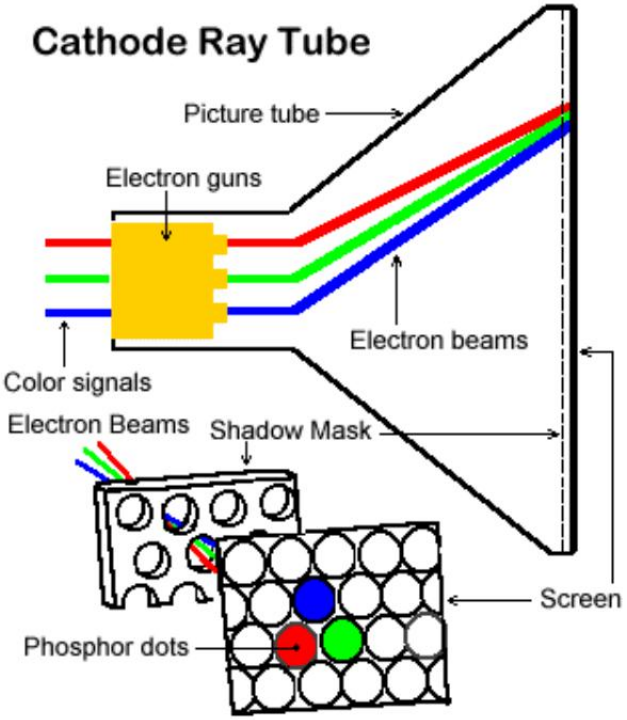
Colors combine by *adding* color spectra



Light *adds* to black.

在黑色背景上叠加光。

Examples of additive color systems 加色系统示例



CRT phosphors

阴极射线管荧光体

multiple projectors

多台投影机

Superposition 叠加原理

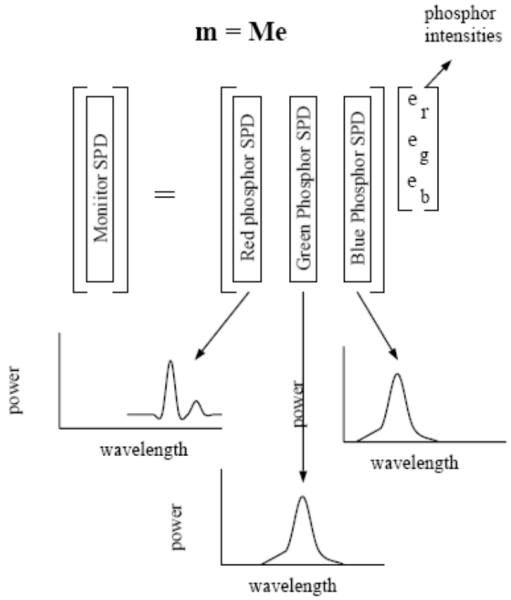


Figure from B. Wandell, 1996

Additive mixing:

The spectral power distribution of the mixture is the sum of the spectral power distributions of the components.

加色混合

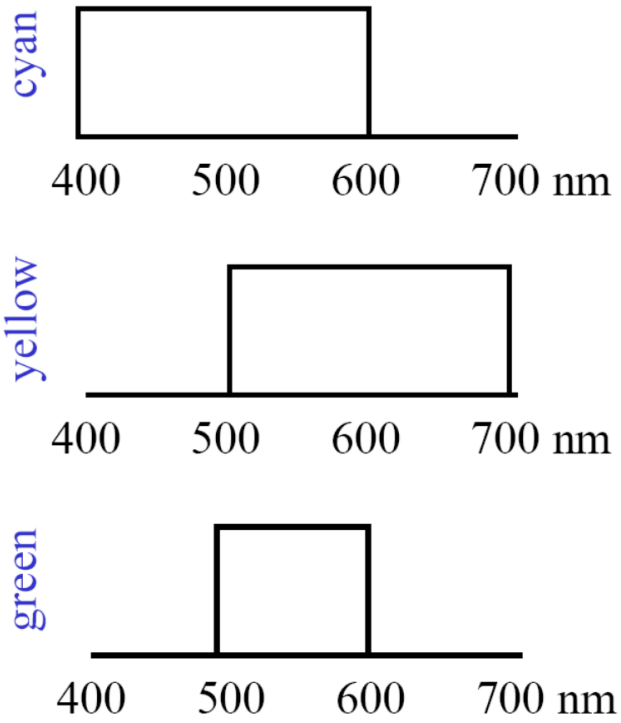
混合光的光谱功率分布等于各组成光的光谱功率分布之和。

m 为显示器的光谱功率分布。

M 为红、绿、蓝荧光体的光谱功率分布矩阵。

e 为荧光体强度。图中坐标为功率与波长。

Subtractive color mixing 减色混合



Colors combine by *multiplying* color spectra.

Pigments *remove* color from incident light (white).

各颜色的光谱相乘，形成混合颜色。颜料从入射白光中吸收部分颜色成分。图中青色与黄色混合形成绿色。

Examples of subtractive color systems 减色系统示例

Printing on paper

Crayons

Most photographic film

纸张印刷

蜡笔

大多数摄影胶片

Standard color spaces 通用颜色空间

Use a common set of primaries/color matching functions

Linear color space examples

- RGB
- CIE XYZ

Non-linear color space

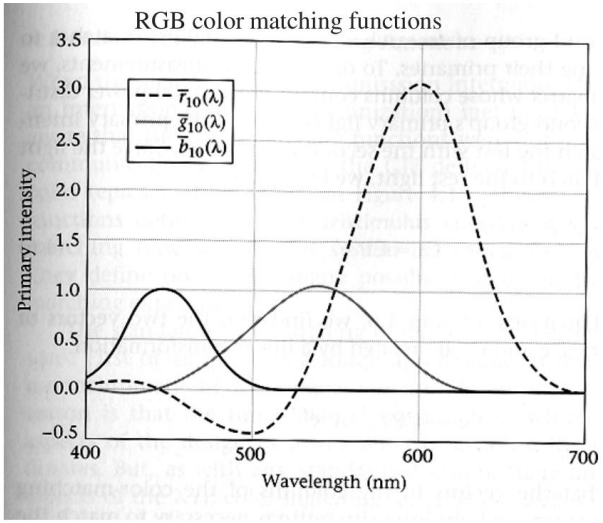
- HSV

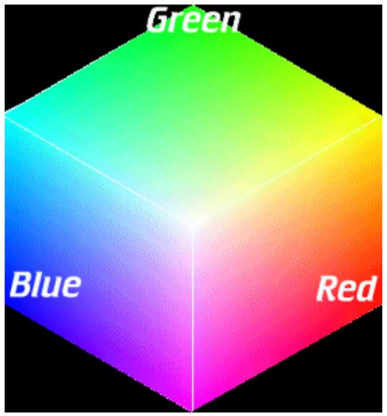
采用共同的一组基色与颜色匹配函数。
线性颜色空间示例为 RGB 和 CIE XYZ。
非线性颜色空间示例为 HSV。

RGB color space　RGB 颜色空间

Single wavelength primaries

Good for devices (e.g., phosphors for monitor), but not for perception





单波长基色。

适合设备表征，例如显示器的荧光体。对人类颜色感知的表征能力有限。

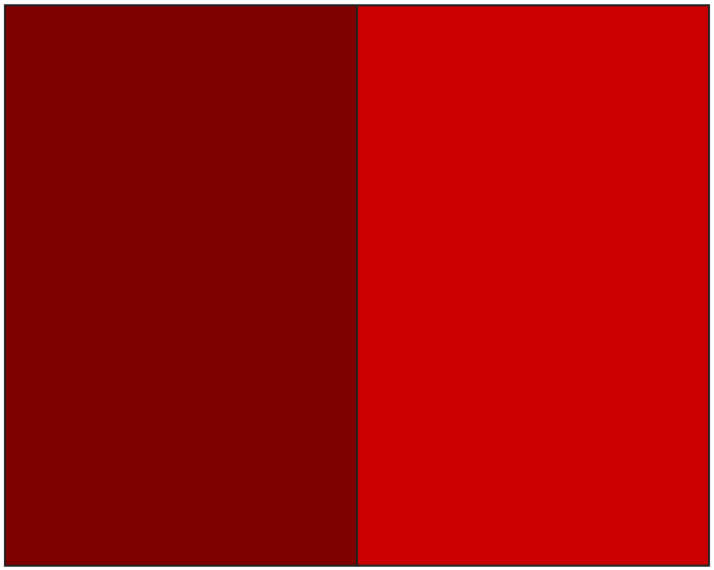
RGB 颜色匹配函数。

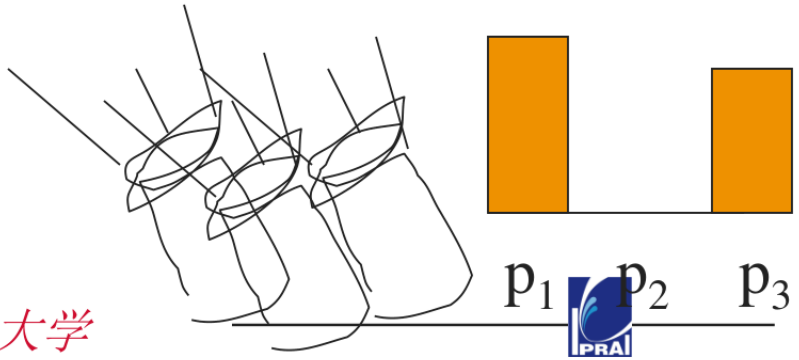
函数图纵轴为基色强度，横轴为波长，单位为纳米。

RGB 立方体的三个坐标轴分别为红、绿、蓝分量。

Color matching experiment 2　颜色匹配实验 2

p_1、p_2、p_3 为三种基色的混合用量。

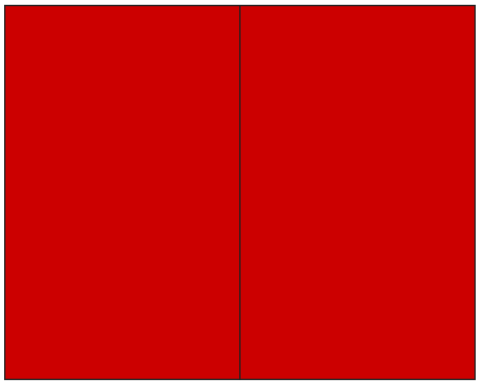


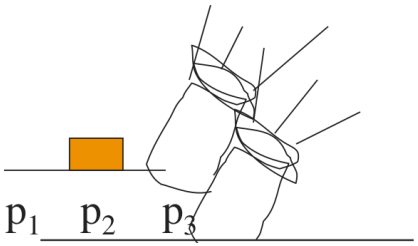


Slide credit: W Freeman

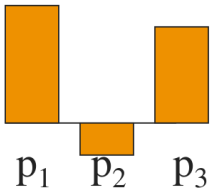
Color matching experiment 2　颜色匹配实验 2

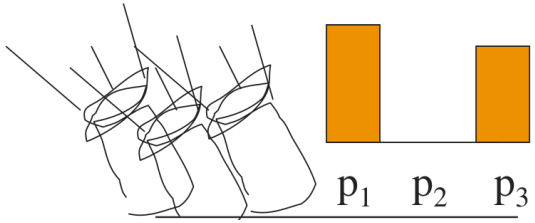
We say a "negative" amount of p_2 was needed to make the match, because we added it to the test color's side.





The primary color amounts needed for a match:





将 p_2 加到测试色一侧后实现颜色匹配，因此把所需的 p_2 用量记为负值。

实现匹配所需的各基色用量。

p_1、p_2、p_3 分别表示三种基色的用量。

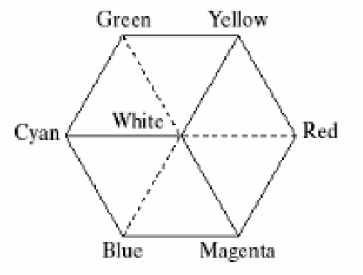
HSV color space　HSV 颜色空间

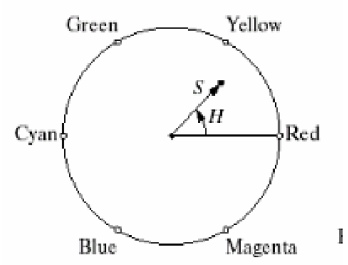
Hue, **S**aturation, **V**alue (Brightness)

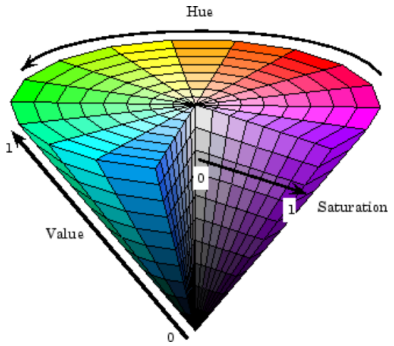
Nonlinear – reflects topology of colors by coding **hue** as an angle

Matlab: `hsv2rgb`, `rgb2hsv`.

OpenCV: cvCvtColor(floatsrc, floathsv, CV_BGR2HSV)







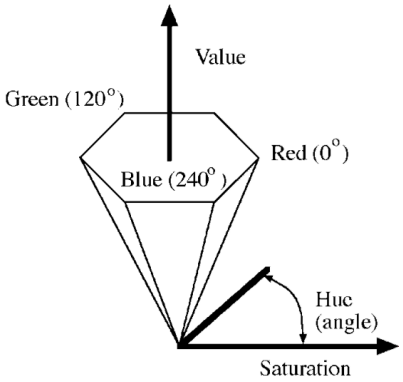


Image from mathworks.com

HSV 分别表示色相、饱和度和明度（亮度）。

该非线性颜色空间以角度编码色相，反映颜色的拓扑结构。

Matlab 颜色转换函数为 hsv2rgb 和 rgb2hsv。

OpenCV 颜色转换使用图中的 cvCvtColor 调用。

H 为色相，S 为饱和度，V 为明度。

色相示例为红色 0°、绿色 120°、蓝色 240°。

颜色空间转换

RGB->HIS

$$\theta = \cos^{-1}\left\{\frac{[(R-G)+(R-B)]/2}{\sqrt{(R-G)^2+(R-B)(G-B)}}\right\}$$

$$H = \begin{cases} \theta & B \le G \\ 360-\theta & B > G \end{cases}$$

$$S = 1 - \frac{3\cdot\min(R,G,B)}{R+G+B}$$

$$I = (R+G+B)/3$$

RGB->HSL,HSV

$$h = \begin{cases} 0^\circ & \text{if } max = min \\ 60^\circ \times \frac{g-b}{max-min} + 0^\circ, & \text{if } max = r \text{ and } g \ge b \\ 60^\circ \times \frac{g-b}{max-min} + 360^\circ, & \text{if } max = r \text{ and } g < b \\ 60^\circ \times \frac{b-r}{max-min} + 120^\circ, & \text{if } max = g \\ 60^\circ \times \frac{r-g}{max-min} + 240^\circ, & \text{if } max = b \end{cases}$$

$$l = \tfrac{1}{2}(max + min)$$

$$s = \begin{cases} 0 & \text{if } l = 0 \text{ or } max = min \\ \frac{max-min}{max+min} = \frac{max-min}{2l}, & \text{if } 0 < l \le \frac{1}{2} \\ \frac{max-min}{2-(max+min)} = \frac{max-min}{2-2l}, & \text{if } l > \frac{1}{2} \end{cases}$$

$$s = \begin{cases} 0, & \text{if } max = 0 \\ \frac{max-min}{max} = 1 - \frac{min}{max}, & \text{otherwise} \end{cases}$$

$$v = max$$

RGB 到 HSI 的转换，图中 HIS 即 HSI。

RGB 到 HSL、HSV 的转换。

H 为色相，S 为饱和度，I 为强度，L 为明度，V 为明度（最大通道值）。

max 与 min 分别为 r、g、b 中的最大值和最小值。

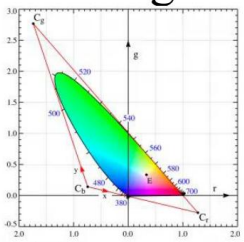
if 为条件成立时，otherwise 为其余情况。

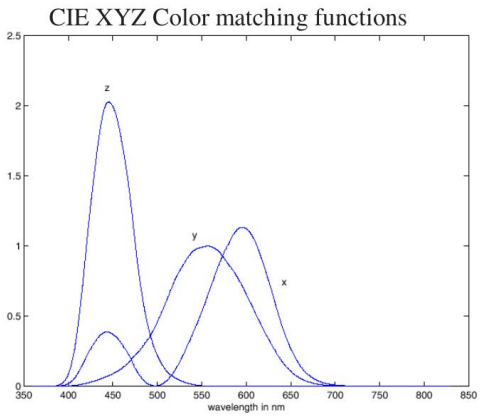
CIE XYZ color space　CIE XYZ 颜色空间

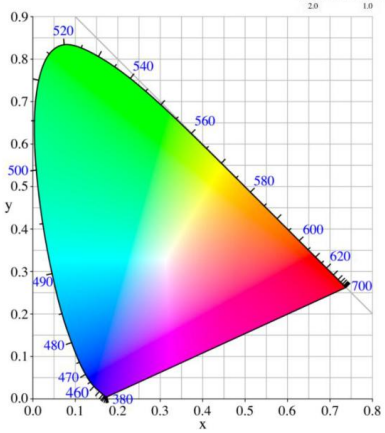
Established by the commission international d'eclairage (CIE), 1931

Usually projected to display:

(x,y) = (X/(X+Y+Z), Y/(X+Y+Z))







由国际照明委员会 CIE 于 1931 年建立。

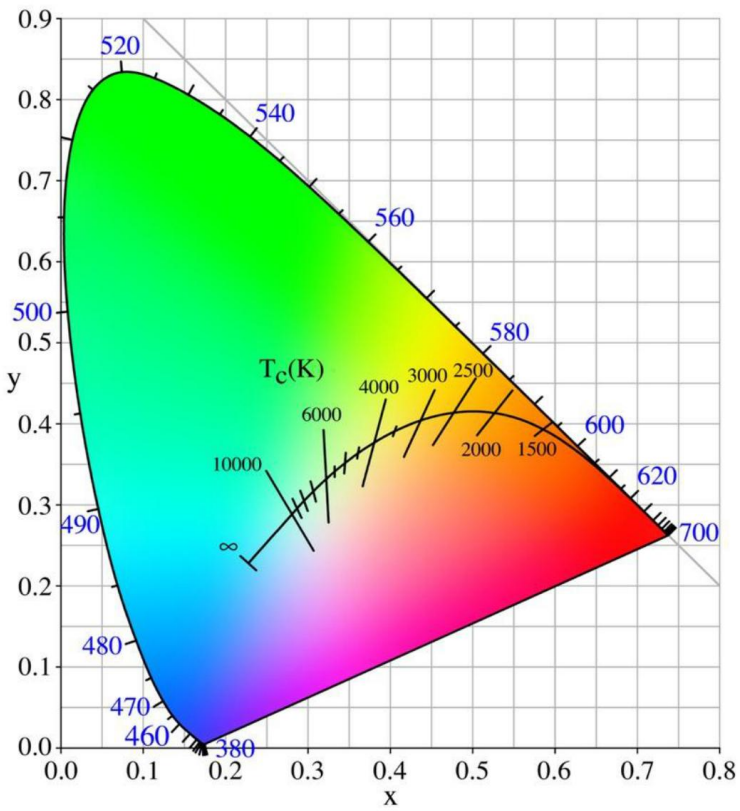
通常投影为二维色度坐标进行显示。

x=X/(X+Y+Z)
y=Y/(X+Y+Z)

CIE XYZ 颜色匹配函数。

函数图横轴为波长，单位为纳米。

色度图与色温轨迹



The figure shows:

- The spectral locus (i.e., the colours of monochromatic lights)
- The black-body locus (i.e., the colors of heated black-bodies)
- Some common illuminants

图中包含光谱轨迹，即各单色光的颜色。

黑体轨迹表示受热黑体的颜色。

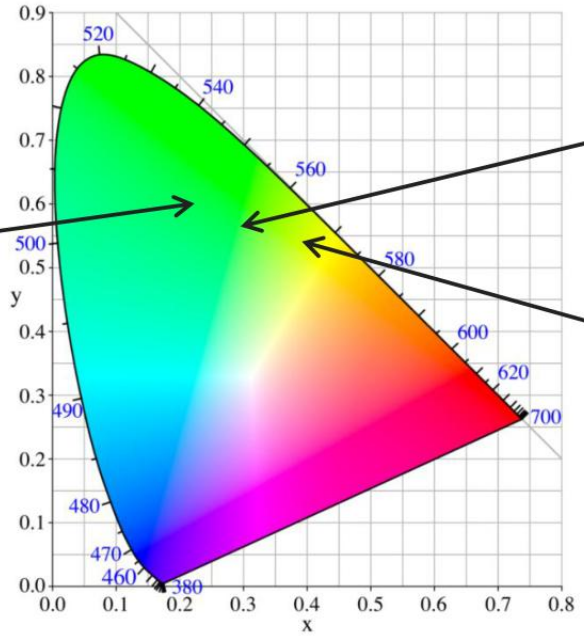
图中同时列出若干常见光源。

$T_{(C)}$ 表示色温，单位为开尔文 K。

Distances in color space 颜色空间距离

Are distances between points in a color space perceptually meaningful?

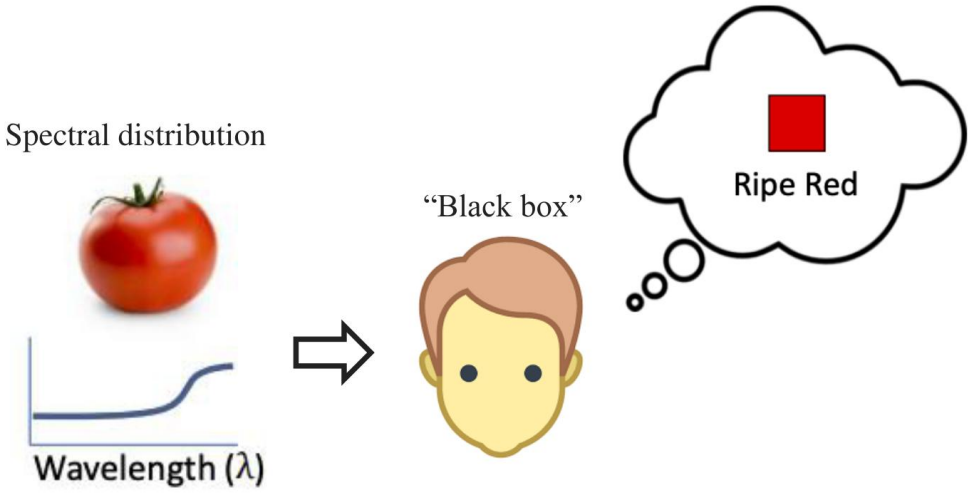
考察颜色空间中两点距离的感知意义。



Quantifying color 颜色量化

Spectral distributions go through a “black box” (human visual system) and are perceived as color

The only way to quantify the “black box” is to perform a human study



Source: M. Brown

光谱分布经过人类视觉系统这一“黑箱”，形成颜色感知。

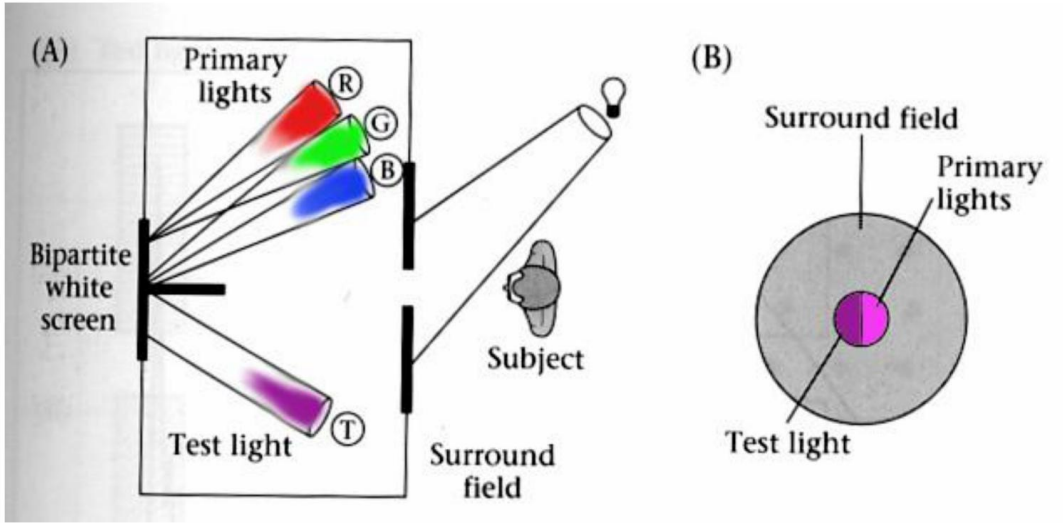
量化这一“黑箱”的唯一途径是开展人类视觉实验。

图中 Spectral distribution 为光谱分布，Black box 为黑箱。

Ripe Red 表示成熟果实的红色，Wavelength 表示波长。

Color matching experiments 颜色匹配实验

We would like to understand which spectra produce the same color sensation in people under similar viewing conditions



Wandell, Foundations of Vision, 1995

研究在相近观察条件下，哪些光谱会使人产生相同的颜色感觉。

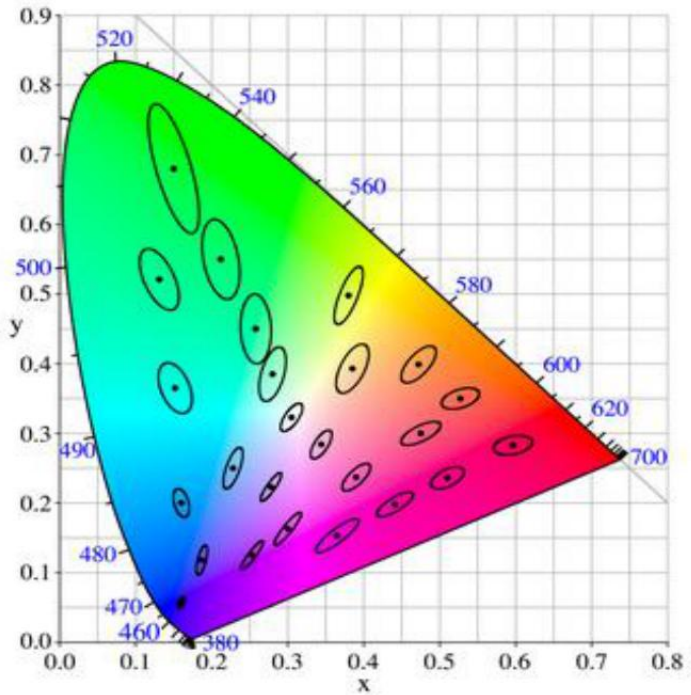
Primary lights 为基色光，Test light 为测试光。

Bipartite white screen 为分成两部分的白色屏幕。

Subject 为受试者，Surround field 为周围视场。

Distances in color space 颜色空间距离

Not necessarily: CIE XYZ is **not** a *uniform* color space, so magnitude of differences in coordinates are poor indicator of color "distance".



CIE XYZ 的均匀性有限，因此坐标差值的大小难以充分反映感知颜色距离。

麦克亚当椭圆表示颜色的最小可觉差。

McAdam ellipses:
Just noticeable differences in color

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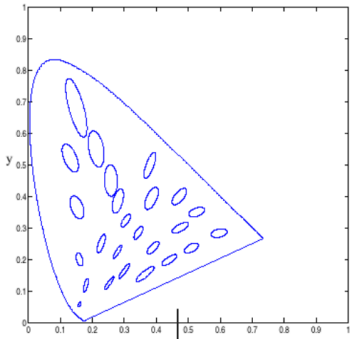
Uniform color spaces 均匀颜色空间

Attempt to correct this limitation by remapping color space so that just-noticeable differences are contained by circles → distances more perceptually meaningful.

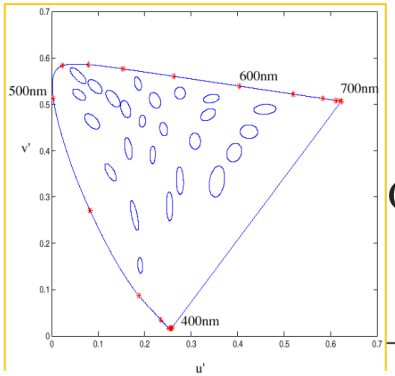
Examples:

CIE u'v'

CIE Lab



CIE XYZ



CIE u'v'

通过重新映射颜色空间，使最小可觉差由圆形区域表示，从而让距离更能反映感知差异。

示例为 CIE u′v′ 和 CIE Lab。

右上图为 CIE XYZ，右下图为 CIE u′v′。

Opponent color and chromatism 对立色与色度

Lab:

L: Luminance (0,100)

a: green←→ red (-127,127)

b: blue←→yellow (-127,127)

$$\begin{cases} R = \text{gamma}(\frac{r}{255.0}) \\ G = \text{gamma}(\frac{g}{255.0}) \\ B = \text{gamma}(\frac{b}{255.0}) \end{cases} \qquad \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = M * \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

（1）

$$\text{gamma}(x) = \begin{cases} \left(\frac{x + 0.055}{1.055}\right)^{2.4} & (x > 0.04045) \\ \frac{x}{12.92} & (\text{其它}) \end{cases}$$

（2）XYZ转LAB

$$L^\star = 116 f(Y/Y_n) - 16$$
$$a^\star = 500[f(X/X_n) - f(Y/Y_n)]$$
$$b^\star = 200[f(Y/Y_n) - f(Z/Z_n)]$$

$$f(t) = \begin{cases} t^{1/3} & \text{if } t > (\frac{6}{29})^3 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29} & \text{otherwise} \end{cases}$$

M=

0.4124,0.3576,0.1805

0.2126,0.7152,0.0722

0.0193,0.1192,0.9505

等同于如下公式：

X = var_R * 0.4124 + var_G * 0.3576 + var_B * 0.1805
Y = var_R * 0.2126 + var_G * 0.7152 + var_B * 0.0722
Z = var_R * 0.0193 + var_G * 0.1192 + var_B * 0.9505

Lab 颜色空间。

L 为明度，范围为 0 至 100。

a 为绿色与红色对立分量，范围为 −127 至 127。

b 为蓝色与黄色对立分量，范围为 −127 至 127。

gamma 表示伽马函数，f 表示分段函数。

if 为条件成立时，otherwise 为其余情况。

Opponent color and chromatism 对立色与色度

Lab:

L: Luminance (0,100)

a: green←→ red (-127,127)

b: blue←→yellow (-127,127)

YUV:

Y=0.3R+0.59G+0.11B

U: B-Y

V: R-Y

Lab 颜色空间
L 为明度，范围为 0 至 100。
a 为绿色与红色对立分量，范围为 −127 至 127。
b 为蓝色与黄色对立分量，范围为 −127 至 127。

YUV 颜色空间
Y 为亮度，Y=0.3R+0.59G+0.11B。
U 为蓝色色差，U=B−Y。
V 为红色色差，V=R−Y。

Color perception 颜色感知

Color/lightness constancy

The ability of the human visual system to perceive the intrinsic reflectance properties of the surfaces despite changes in illumination conditions

颜色与明度恒常性

即使光照条件发生变化，人类视觉系统仍能感知表面固有的反射属性。

萨金特《博伊特的女儿们》，1882 年。

J. S. Sargent, The Daughters of Edward D. Boit, 1882

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Chromatic adaptation 色适应

The visual system changes its sensitivity depending on the luminances prevailing in the visual field

- The exact mechanism is poorly understood

Adapting to different brightness levels

- Changing the size of the iris opening (i.e., the aperture) changes the amount of light that can enter the eye
- Think of walking into a building from full sunshine

Adapting to different color temperature

- The receptive cells on the retina change their sensitivity
- For example: if there is an increased amount of red light, the cells receptive to red decrease their sensitivity until the scene looks white again
- We actually adapt better in brighter scenes: This is why candlelit scenes still look yellow

http://www.schorsch.com/kbase/glossary/adaptation.html

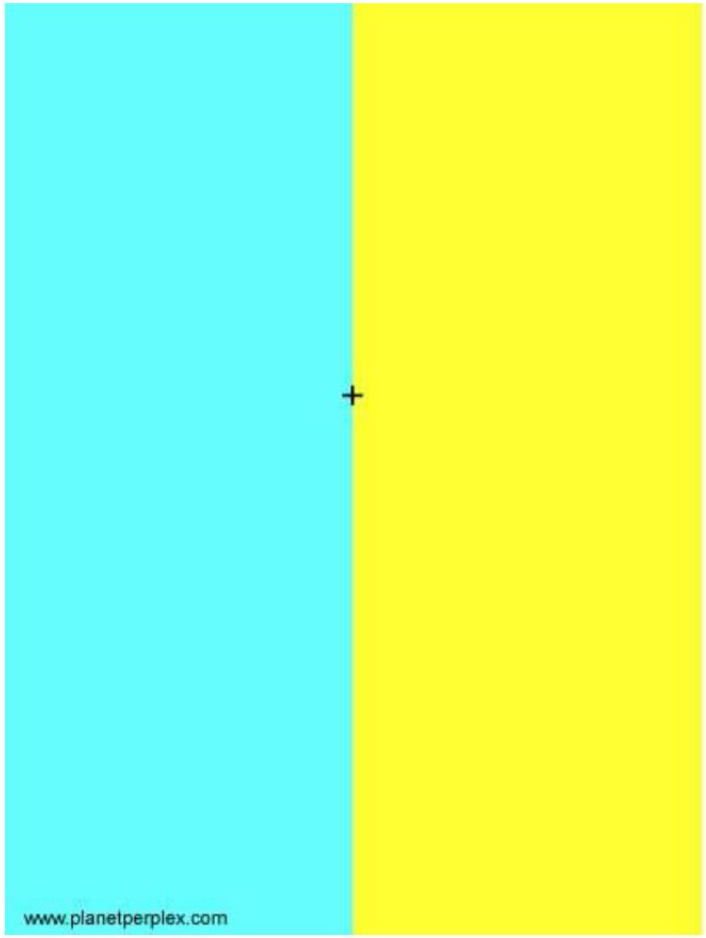
视觉系统依据视野内的主要亮度水平调节自身敏感度。具体机制仍有待深入理解。

各类亮度水平的适应。虹膜开口即光圈的大小变化，会改变进入眼内的光量。例如从明亮阳光下走入建筑内部。

各类色温的适应。视网膜感光细胞会改变自身敏感度。例如红光增多时，对红光敏感的细胞会降低敏感度，直至场景重新呈现白色。

明亮场景中的适应效果较好，因此烛光场景仍会呈黄色。

Chromatic adaptation 色适应



色彩平衡、光源估计与适应。

Color balance, Illuminant estimation and adaptation

What color is the dress? 裙装颜色感知

White-Golden

白色与金色

or Blue-Black?

或蓝色与黑色

https://www.wired.com/2015/02/science-one-agrees-color-dress/

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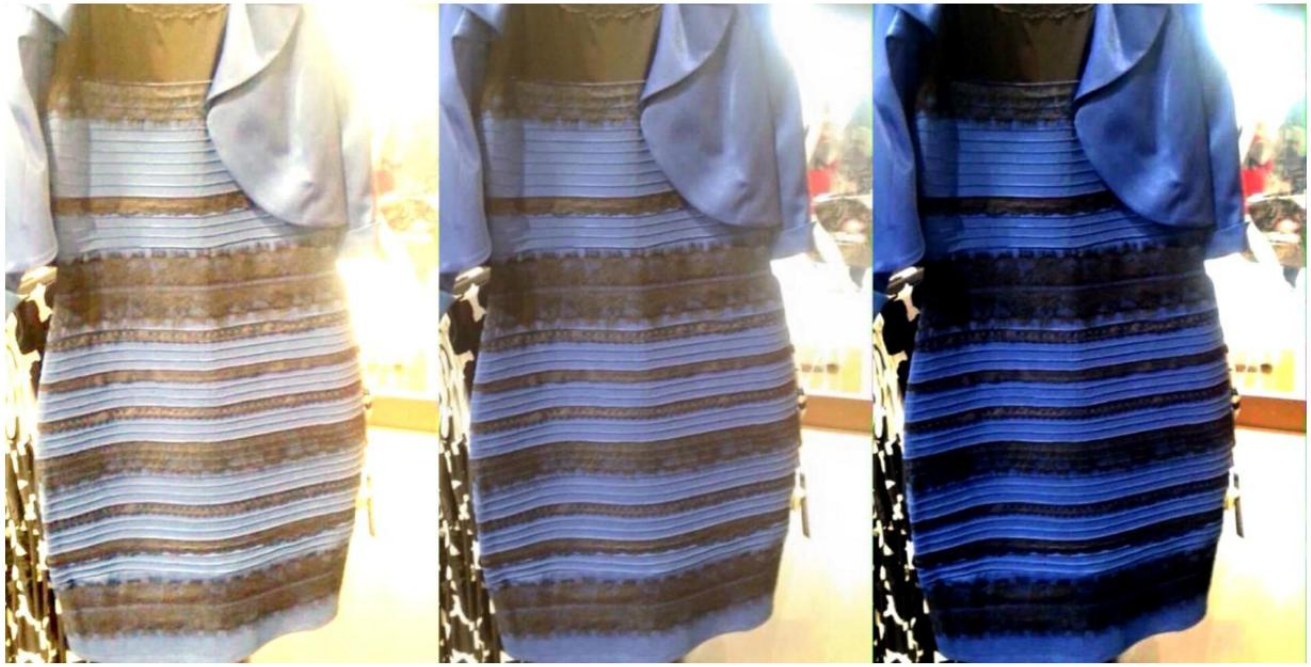
What color is the dress? 裙装颜色感知



White-Golden

白色与金色

or Blue-Black?

或蓝色与黑色

https://www.wired.com/2015/02/science-one-agrees-color-dress/

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This strawberry cake has no red pixels!　草莓蛋糕的颜色错觉

这幅草莓蛋糕图像中的红色像素数量为零。

https://www.digitaltrends.com/photography/non-red-strawberries/

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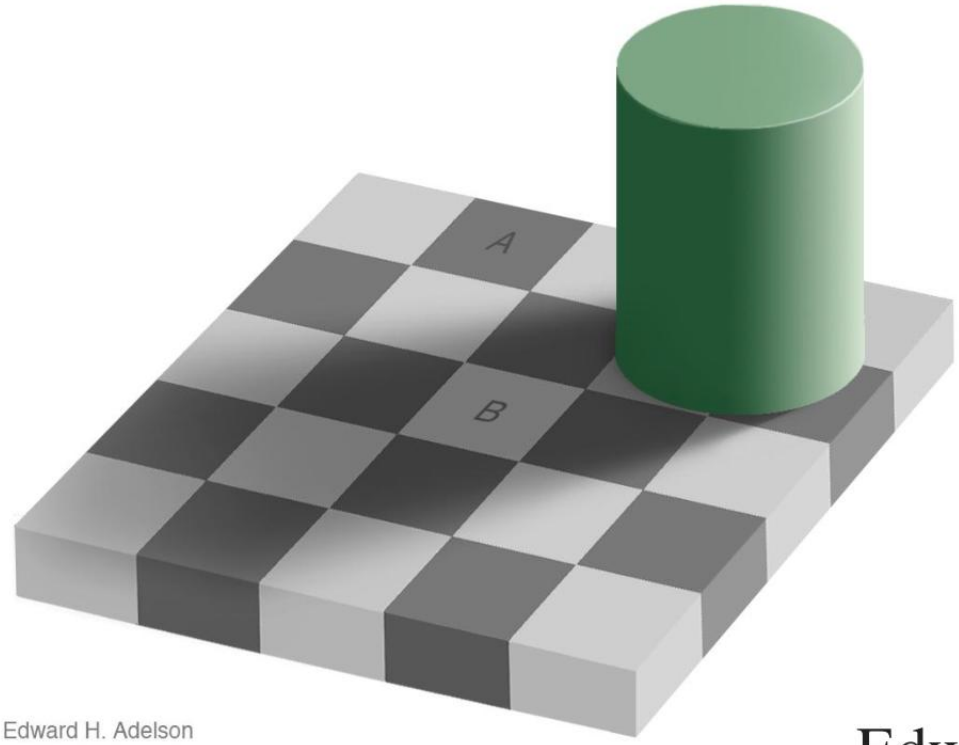
Brightness perception 亮度感知



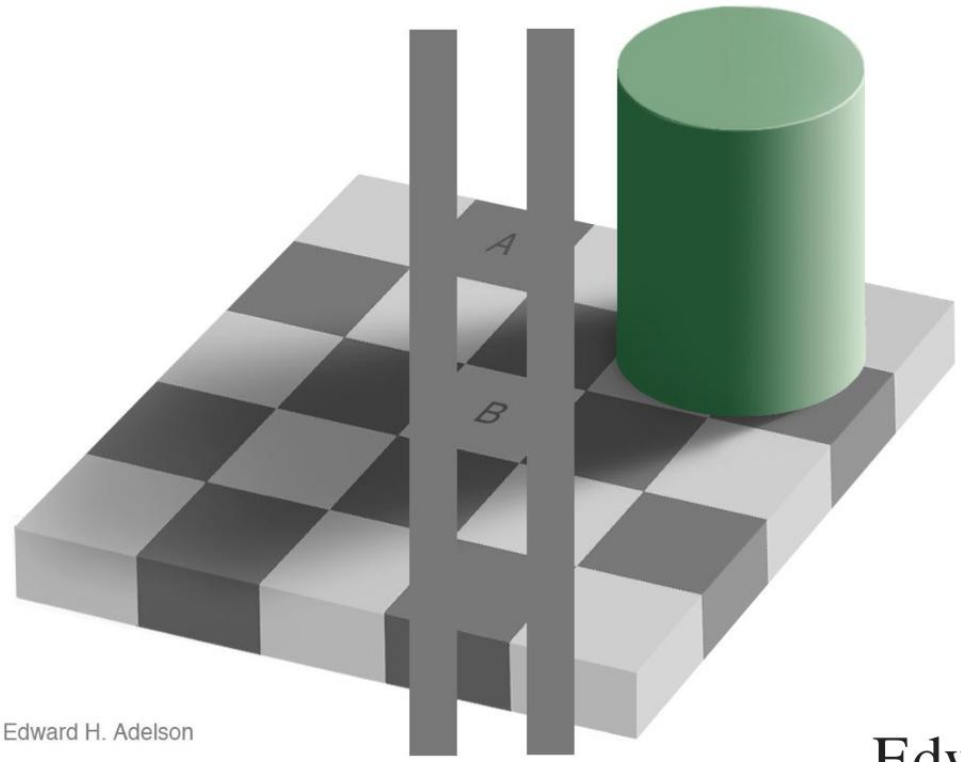
Edward H. Adelson

Edward Adelson

Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h

亮度感知

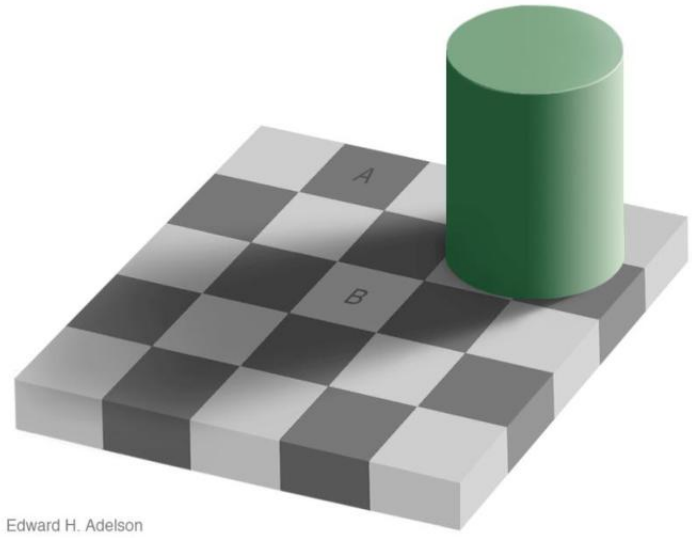


Edward Adelson

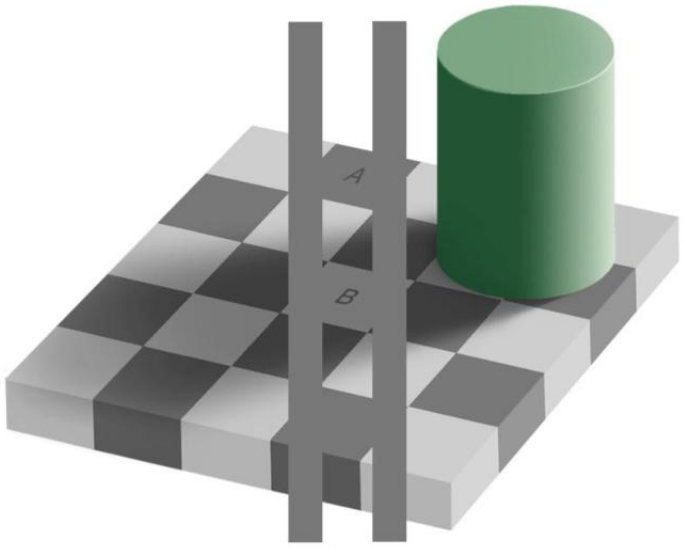
Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h

亮度感知



Edward H. Adelson



Edward Adelson

Edward Adelson

http://web.mit.edu/persci/people/adelson/illusions_demos.h

Color constancy 颜色恒常性

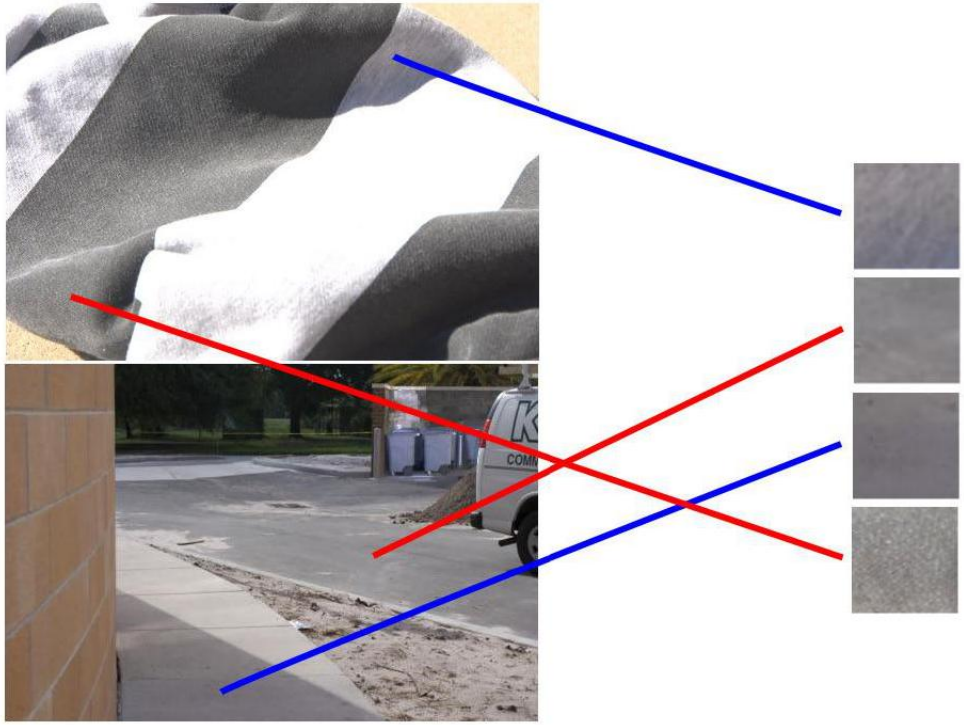
Interpret surface in terms of albedo or “true color”, rather than observed intensity

- Humans are good at it
- Computers are not nearly as good

表面解释采用反照率或“固有颜色”，并与观测强度区分。

人类擅长这种解释。

计算机在这方面的表现与人类仍有较大差距。



White balance 白平衡

Analogous to color constancy mechanisms in human vision, cameras have mechanisms to adapt to the illumination in the environment so that neutral (white or gray) objects look neutral

Incorrect white balance

Correct white balance

http://www.cambridgeincolour.com/tutorials/white-balance.htm

相机具有适应环境光照的机制，与人类视觉中的颜色恒常性机制相似，使白色或灰色的中性物体仍呈中性色。左图白平衡失调，右图白平衡适当。

White balance 白平衡

Film cameras:

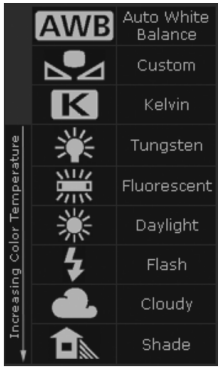
Different types of film or different filters for different illumination conditions

Digital cameras:

Automatic white balance

White balance settings corresponding to several common illuminants

Custom white balance using a reference object



http://www.cambridgeincolour.com/tutorials/white-balance.htm

胶片相机。各类光照条件采用相应胶片或滤光片。

数码相机。自动白平衡、对应若干常见光源的白平衡设置，以及利用参考物体自定义白平衡。

图中设置包括自动、自定义、开尔文色温、钨丝灯、荧光灯、日光、闪光灯、阴天与阴影。

图中箭头表示色温逐渐升高。

White balance 白平衡

Von Kries adaptation: Multiply each channel by a *gain factor*

Best way: gray card

Take a picture of a neutral object (white or gray)

If the object is recorded as r_w, g_w, b_w use weights $1/r_w$, $1/g_w$, $1/b_w$

von Kries 色适应
各通道乘以一个增益因子。

最佳方式为灰卡。
拍摄白色或灰色的中性物体。

若物体记录值为 r_w、g_w、b_w，则采用权重 1/r_w、1/g_w、1/b_w。

White balance 白平衡

Without gray cards: we need to "guess" which pixels correspond to white objects

Gray world assumption

- The image average r_{ave}, g_{ave}, b_{ave} is gray
- Use weights $1/r_{ave}$, $1/g_{ave}$, $1/b_{ave}$

Brightest pixel assumption

- Highlights usually have the color of the light source
- Use weights inversely proportional to the values of the brightest pixels

Gamut mapping

- Gamut: convex hull of all pixel colors in an image
- Find the transformation that matches the gamut of the image to the gamut of a "typical" image under white light

Use image statistics, learning techniques

缺少灰卡时，需要估计哪些像素对应白色物体。

灰度世界假设。图像均值 r_ave、g_ave、b_ave 对应灰色，采用权重 1/r_ave、1/g_ave、1/b_ave。

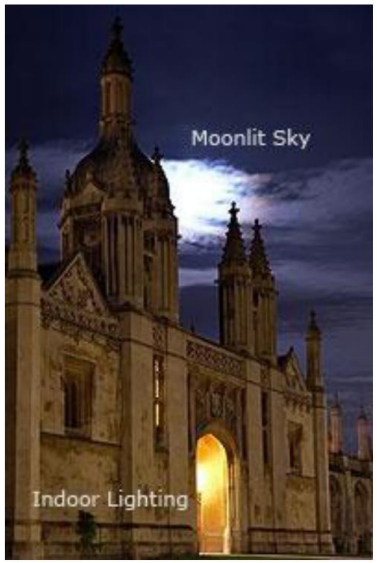
最亮像素假设。高光通常呈光源颜色，权重与最亮像素值成反比。

色域映射。色域为图像中全部像素颜色的凸包。求取一种变换，使图像色域与白光下“典型”图像的色域匹配。

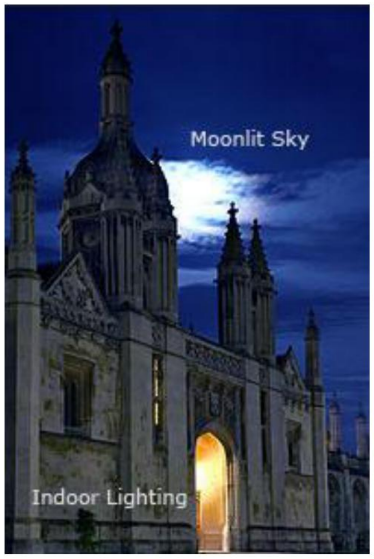
利用图像统计或学习方法。

Mixed illumination 混合光照

When there are several types of illuminants in the scene, different reference points will yield different results



Reference: moon

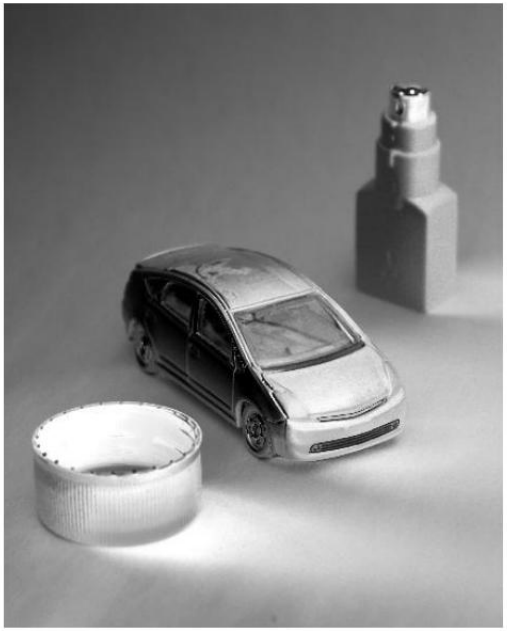


Reference: stone

http://www.cambridgeincolour.com/tutorials/white-balance.htm

场景中存在多类光源时，各参考点会产生相应的白平衡差异。左图以月亮为参考，右图以石块为参考。Moonlit Sky 为月光天空，Indoor Lighting 为室内照明。

Spatially varying white balance　空间变化白平衡



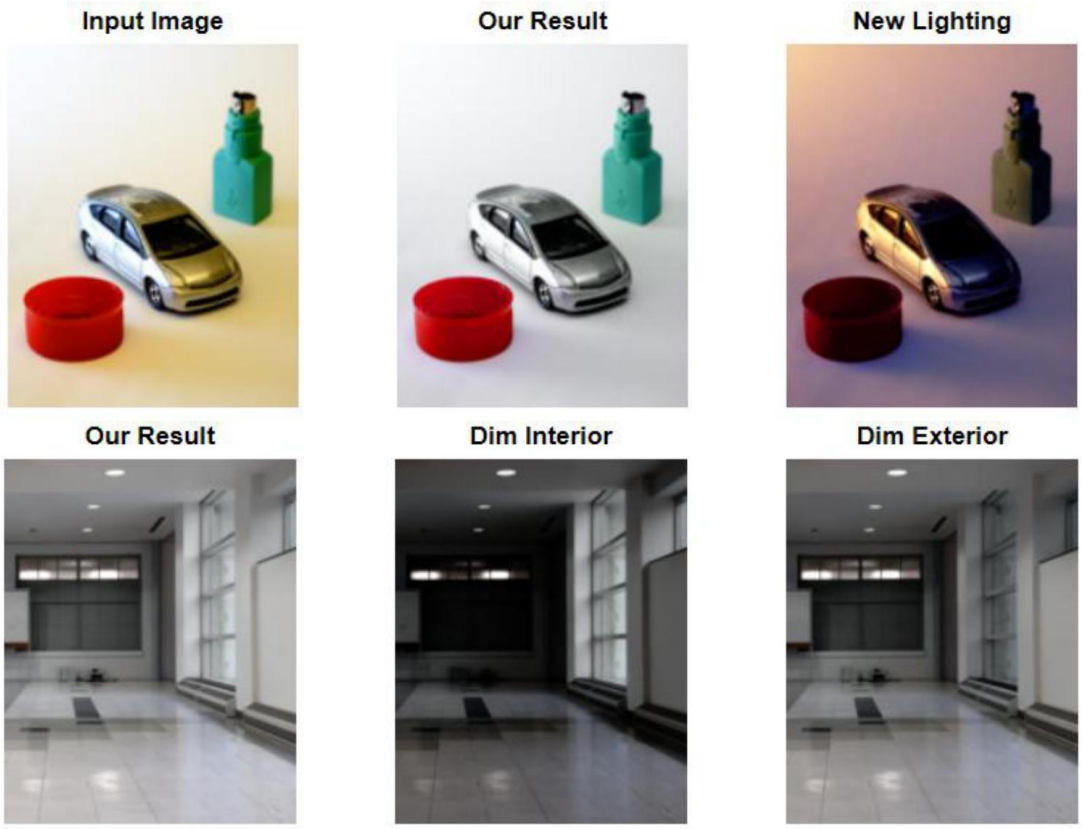
Input　　Alpha map　　Output

输入　　混合权重图　　输出

《用于空间变化白平衡的混合光照估计》，SIGGRAPH 2008。

E. Hsu, T. Mertens, S. Paris, S. Avidan, and F. Durand, Light Mixture Estimation for Spatially Varying White Balance, SIGGRAPH 2008

空间变化白平衡示例



上排从左至右为输入图像、本方法结果与新光照。下排从左至右为本方法结果、室内区域较暗与室外区域较暗。

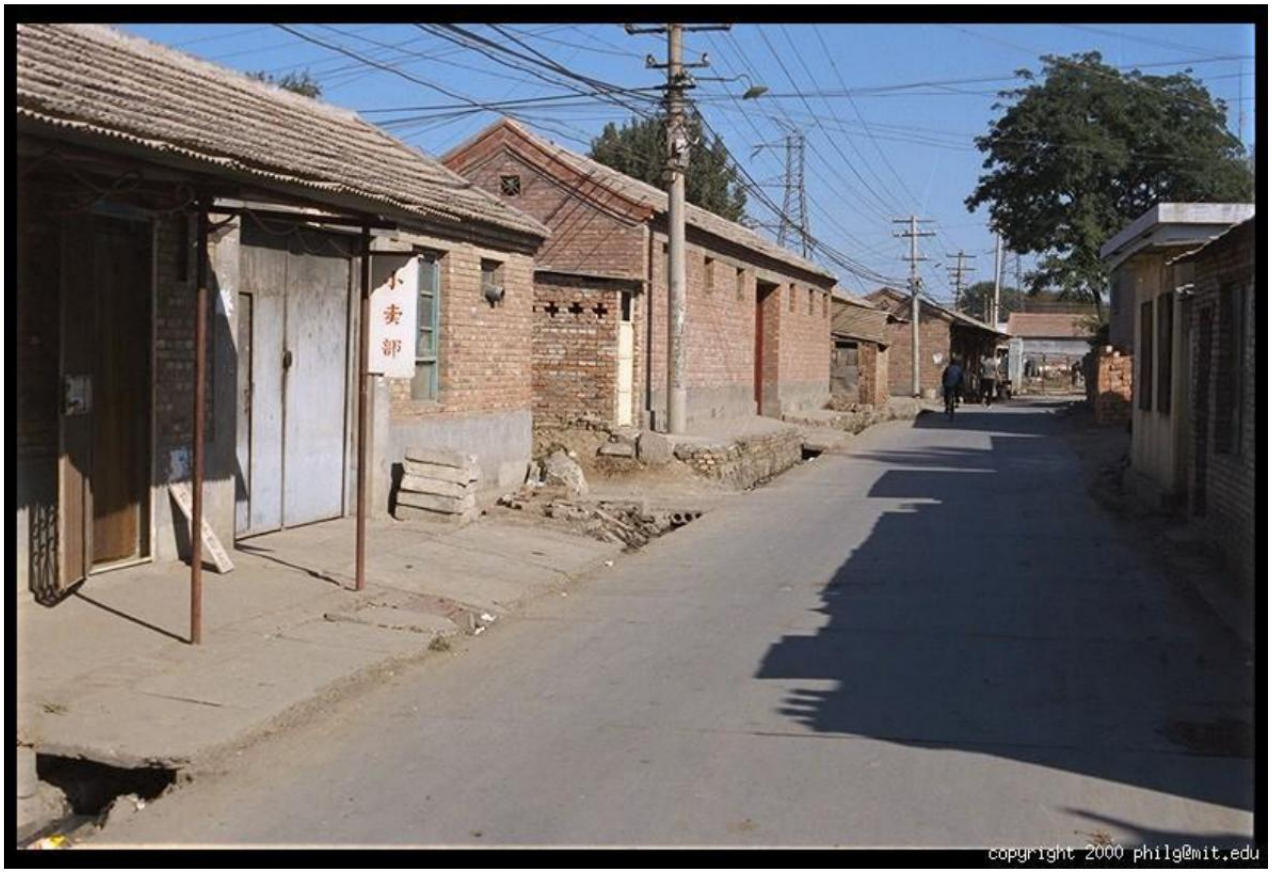
Color cues for image understanding 图像理解中的颜色线索

Recognition 识别

Segmentation 分割

Most information in intensity　强度信息与颜色信息

图像的大部分信息由强度承载。



原始图像

Original image

Most information in intensity　强度信息与颜色信息

图像的大部分信息由强度承载。

仅显示颜色，强度保持恒定。

Only color shown – constant intensity

Most information in intensity　强度信息与颜色信息

图像的大部分信息由强度承载。

仅显示强度，颜色保持恒定。

Only intensity shown – constant color